Exact Second-Order Asymptotics for the Gaussian Lossy Gray–Wyner Problem

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Abstract

We characterize second-order rates at Pareto-minimal boundary tuples of the bivariate Gaussian Gray–Wyner problem under quadratic distortion and a joint excess-distortion probability constraint, for nonsingular correlated sources and positive distortions below the marginal variances. For nonzero common rate, we develop a covariance-shell covering lemma that supplies the missing achievability counterpart to the converse of Liu, Courtade, Cuff, and Verdú [IEEE Trans. Inf. Theory, 2020]. The bounds match at boundary points where increasing either private rate can reduce the required common rate; scalar Gaussian converses complete the remaining one-source cases. The covering lemma replaces discrete type covering by a common and two private coverings of an entire empirical covariance shell, with negligible second-order overhead. Our second contribution is an exact characterization at zero first-order common rate, for which we prove both a new achievability specialization and a joint converse. A common description can still carry information on the square-root blocklength scale, and the two receivers' excess-distortion events remain coupled. The resulting region is a nonlinear bivariate Gaussian probability, rather than two separate point-to-point conditions. The nonzero common rate dispersion depends only on the sum of the two supporting slopes. Numerical evaluations illustrate the gain from a positive second-order common budget and the finite-blocklength distribution of the source fluctuation.

Index Terms

Finite blocklength, dispersion, covariance-shell covering, multiterminal source coding, hypercontractivity.

I. Introduction

Shannon's source coding theorem identified entropy as the first-order limit for lossless compression [1]. Strassen [2] subsequently characterized the leading square-root blocklength correction at a nonvanishing error probability. For lossy compression, Shannon's rate-distortion theory [3] quantified the first-order rate reduction permitted by an average distortion constraint. Ingber and Kochman [4] and Kostina and Verdú [5], [6] developed dispersion and finite-blocklength results, including quadratic Gaussian sources. These refinements quantify the rate needed beyond the first-order limit when blocklength and excess-distortion probability are prescribed.

For two receivers requesting correlated signals, entirely private descriptions forgo shared information, whereas a wholly common description can force each receiver to obtain information needed only by the other. Gray–Wyner coding [7] isolates this allocation problem through one common and two private messages. Gray and Wyner characterized its first-order region using a common auxiliary and two conditional rate-distortion functions. Watanabe [8] obtained a discrete almost-lossless second-order region, and Zhou, Tan, and Motani [9] established the discrete lossy counterpart. For Gaussian sources, the common/private tradeoff and Gaussian optimality were studied in [10], [11]; Yu [12, Theorem 3 and Corollary 2] gave an explicit boundary. Liu, Courtade, Cuff, and Verdú [11, Theorem 17] also proved a second-order converse for positive-weight supporting directions.

The missing direct argument must replace finite-alphabet type covering by a construction for an entire Gaussian covariance shell. One common codeword must simultaneously leave two residual vectors compatible with the private budgets, uniformly even when the common Gaussian component loses rank. We establish this covering result and match the Gaussian converse, with scalar converses completing the one-source directions. We also characterize the zero common rate corner. There, a common description may still contain order-$\sqrt{n}$ information, and separate fixed-direction converse bounds do not retain the joint fluctuation needed for an exact region.

At nonzero common rate, our covariance-shell lemma supplies one common and two private coverings at the Gaussian design rates with redundancy negligible on the second-order scale. The subsequent proof parallels the discrete method: the empirical covariance replaces the joint type, Yu's local boundary is expanded, and a scalar central limit theorem yields the achievable offsets. The direct bound matches the positive-weight converse of [11] or a scalar Gaussian converse according to the supporting direction. A complementary-slackness argument further reduces the dispersion to an explicit function of the sum of the two private-rate slopes.

At zero first-order common rate, we prove both the direct specialization and a uniform joint covariance-cell converse for arbitrary codes. The converse bounds the probability of simultaneous successful decoding within each empirical covariance cell, with a polynomial factor independent of the code and of the optimized supporting direction. This finite-blocklength estimate retains the joint event before the Gaussian limit is taken. Gaussian hypercontractivity produces the same nonlinear allocation cost as the directional derivative in the direct argument, giving an exact bivariate Gaussian probability. At zero second-order common rate, private coverings alone attain the full slice, so its coupling comes from the source correlation and joint error criterion rather than a need for common communication.

For distributed lossless compression, Tan and Kosut [13] studied Slepian–Wolf dispersion, and Nomura and Han [14] treated non-mixed and mixed sources by information-spectrum methods. Gaussian successive refinement was studied by No, Ingber, and Weissman [15]; discrete and Gaussian second-order regions and moderate deviations were obtained in [16]. We use the same Euclidean covering idea for the private descriptions, retaining a uniform proof when the radius ratio approaches one. It does not supply the simultaneous common covering required here.

Second-order achievable bounds for coding with side information were developed in [17], [18]. The discrete Wyner–Ziv analysis of [19] combined a type-dependent construction with a matching converse under its stated conditions. Other refinements concern the Kaspi and Fu–Yeung problems [20] and successive refinement with prescribed Gaussian codebooks [21]; these have different observations, constraints, or admissible codebooks. The monograph [22] surveys these models. We distinguish their achievable bounds from exact operational characterizations.

Section II gives the model, existing first-order result, and second-order criterion. Section III states and interprets the theorem and numerical examples. Sections IV and V give the direct and converse proofs, with all code maps specified before analysis. Section VI concludes; the appendices contain the supporting proofs.

Notation. We use the source, description, and rate notation of the discrete lossy Gray–Wyner paper [9, Sections II–III]. Random variables and their realizations use upper- and lower-case letters. The source pair is (X, Y), its distribution is P_{XY}, the common auxiliary is W, and the transmitted messages are S_0, S_1, S_2. Bold symbols are reserved for the vectors and matrices needed in the Gaussian analysis. A source pair and its block may therefore also be written as $\mathbf{X} = (X, Y)^{\mathsf{T}}$ and $\mathbf{X}^n = [\mathbf{X}_1 \ \cdots \ \mathbf{X}_n]$. These are vector representations of the same source, not additional random variables.

Let $\mathbb{N} := \{1, 2, \ldots\}$ and $\mathbb{R}_+ := [0, \infty)$. For integers a, b, let $[a : b] := \{a, a+1, \ldots, b\}$, with $[a : b] = \emptyset$ when $a > b$. Intervals with a comma, such as $[0, 1]$, are real intervals. For $d \in \mathbb{N}$, the spaces $\mathbb{S}^d$, $\mathbb{S}^d_+$, and $\mathbb{S}^d_{++}$ contain the real symmetric, positive-semidefinite, and positive-definite $d \times d$ matrices. For $(\mathbf{A}, \mathbf{B}) \in (\mathbb{S}^d)^2$, write $\mathbf{A} \preceq \mathbf{B}$ when $\mathbf{B} - \mathbf{A} \in \mathbb{S}^d_+$. Write $|\mathbf{A}|$ for a determinant, $\mathbf{I}_d$ for the identity matrix, and $\mathbf{e}_j$ for a coordinate vector of the specified dimension. The norm is Euclidean for vectors and Frobenius for matrices, and $\langle \mathbf{A}, \mathbf{B} \rangle := \operatorname{tr}(\mathbf{A}^{\mathsf{T}}\mathbf{B})$ for matrices of equal dimension. For $\boldsymbol{\mu} \in \mathbb{R}^d$ and $r \in \mathbb{R}_+$, define $\mathcal{B}_d(\boldsymbol{\mu}, r) := \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x} - \boldsymbol{\mu}\| \le r\}$. For a vector $\mathbf{v}$, $[\mathbf{v}]_j$ is its jth coordinate; for a block matrix $\mathbf{x}^n$, $(\mathbf{x}^n)_j$ is its jth row.

The symbols I_Q and h_Q denote mutual information and differential entropy under a specified distribution Q. We write $\mathsf{E}_Q[\cdot]$, $\mathsf{Var}_Q[\cdot]$, and $\mathsf{Cov}_Q(\cdot, \cdot)$ for expectation, variance, and covariance under that distribution. An overline denotes closure, and $\mathbb{1}_{\mathcal{A}}$ is the indicator of an event $\mathcal{A}$. For $\mathbf{S} \in \mathbb{S}^d_+$, $\mathcal{N}(\boldsymbol{\mu}, \mathbf{S})$ denotes the Gaussian distribution, allowing singular covariance. The symbols tr, diag, and rank denote trace, diagonal matrix, and rank. Let Φ be the standard Gaussian cumulative distribution function (CDF), and let $\mathrm{Q} := 1 - \Phi$ be its complementary cumulative distribution function (CCDF). Set $[x]^+ := \max\{x, 0\}$ for $x \in \mathbb{R}$.

We retain natural logarithms and $\exp(x) = e^x$ throughout this Gaussian analysis; rates and offsets are in nats per source pair. The discrete reference used base-two logarithms. Division by $\log 2$ converts a rate, an offset, or an information quantity here to bits; division by $(\log 2)^2$ converts a variance. The supporting slopes are unchanged. Deterministic $O(\cdot)$ and $o(\cdot)$ estimates have the uniformity stated locally, and every probabilistic limit specifies its generating distribution.

II. Problem Formulation and Existing Results

A. Model and Operational Definitions

Let $\mathcal{X} = \mathcal{Y} = \hat{\mathcal{X}} = \hat{\mathcal{Y}} = \mathbb{R}$. Fix $\sigma_X, \sigma_Y > 0$, $\rho \in (0, 1)$, and distortion levels $0 < D_1 < \sigma_X^2$ and $0 < D_2 < \sigma_Y^2$. The source pair (X, Y) has distribution

$$P_{XY} := \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \tag{1}$$

$$\boldsymbol{\Sigma} := \begin{bmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{bmatrix} \in \mathbb{S}^2_{++}. \tag{2}$$

Its marginals are denoted by P_X and P_Y. For $n \in \mathbb{N}$, the source consists of independent pairs (X_i, Y_i), $i \in [1 : n]$, with distribution P_{XY}; write $P_{XY}^n := P_{XY}^{\otimes n}$. The encoder observes (X^n, Y^n). For matrix calculations, set $\mathbf{X}_i := (X_i, Y_i)^{\mathsf{T}}$ and $\mathbf{X}^n := [\mathbf{X}_1 \ \cdots \ \mathbf{X}_n]$. The two rows of a realization $\mathbf{x}^n$ are x^n and y^n. We identify this matrix with the pair (x^n, y^n) when evaluating an encoder or a source event. Decoder 1 reconstructs X^n, and Decoder 2 reconstructs Y^n. The covariance and distortions are known to the code designer.

A sign change of one source reduces $-1 < \rho < 0$ to this model. At $\rho = 0$ the sources are independent; at $|\rho| = 1$ they satisfy $Y = \rho(\sigma_Y/\sigma_X)X$ almost surely and the covariance is singular. Their first-order reductions are recalled in Section II-B. The second-order theorem concerns the nonsingular, correlated case $0 < |\rho| < 1$.

For a deterministic block $\mathbf{x}^n \in \mathbb{R}^{2\times n}$, define its empirical second-moment matrix by

$$\hat{\mathbf{S}}(\mathbf{x}^n) := \frac{1}{n}\mathbf{x}^n(\mathbf{x}^n)^{\mathsf{T}}. \tag{3}$$

The random version is $\hat{\boldsymbol{\Sigma}}_n := \hat{\mathbf{S}}(\mathbf{X}^n)$. Since the source mean is known to be zero, this uncentered empirical matrix is the descriptor used by the code. No assertion that an empirical distribution is Gaussian is made.

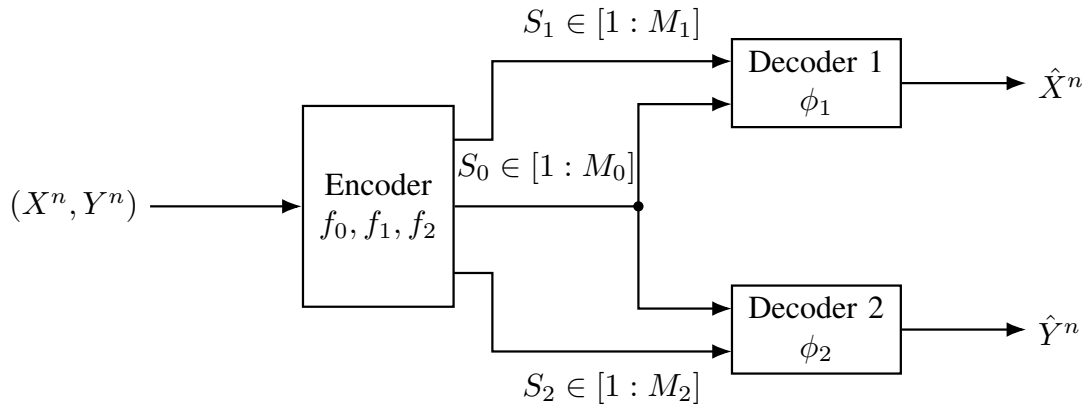


Fig. 1. The Gaussian lossy Gray–Wyner model. The three encoding maps observe (X^n, Y^n). Both decoders receive S_0; Decoder 1 also receives S_1 and reconstructs X^n, while Decoder 2 also receives S_2 and reconstructs Y^n.

The single-letter distortions are $d_X(x, \hat{x}) := (x - \hat{x})^2$ and $d_Y(y, \hat{y}) := (y - \hat{y})^2$. Their block extensions are

$$\begin{aligned} d_X(x^n, \hat{x}^n) &:= \frac{1}{n} \sum_{i \in [1:n]} d_X(x_i, \hat{x}_i) \\ &= \frac{1}{n} \left\| x^n - \hat{x}^n \right\|^2, \end{aligned} \tag{4}$$

$$\begin{aligned} d_Y(y^n, \hat{y}^n) &:= \frac{1}{n} \sum_{i \in [1:n]} d_Y(y_i, \hat{y}_i) \\ &= \frac{1}{n} \left\| y^n - \hat{y}^n \right\|^2. \end{aligned} \tag{5}$$

For $(M_0, M_1, M_2) \in \mathbb{N}^3$, an (n, M_0, M_1, M_2) code consists of three Borel encoding maps and two decoding maps:

$$f_i : \mathcal{X}^n \times \mathcal{Y}^n \longrightarrow [1 : M_i], \qquad i \in [0 : 2], \tag{6}$$

$$\phi_1 : [1 : M_0] \times [1 : M_1] \longrightarrow \hat{\mathcal{X}}^n, \tag{7}$$

$$\phi_2 : [1 : M_0] \times [1 : M_2] \longrightarrow \hat{\mathcal{Y}}^n. \tag{8}$$

The three maps share the same joint observation; they are not separate source encoders. Write $f := (f_0, f_1, f_2)$ for their aggregate and $S_i := f_i(X^n, Y^n)$ for the messages. The random reproductions are $\hat{X}^n := \phi_1(S_0, S_1)$ and $\hat{Y}^n := \phi_2(S_0, S_2)$. For a deterministic input, abbreviate

$$\hat{x}^n(\mathbf{x}^n) := \phi_1(f_0(\mathbf{x}^n), f_1(\mathbf{x}^n)), \tag{9}$$

$$\hat{y}^n(\mathbf{x}^n) := \phi_2(f_0(\mathbf{x}^n), f_2(\mathbf{x}^n)). \tag{10}$$

Define the individual excess-distortion sets, their union, and the joint successful set by

$$\mathcal{E}_{1,n} := \{\mathbf{x}^n \in \mathbb{R}^{2\times n} : d_X(x^n, \hat{x}^n(\mathbf{x}^n)) > D_1\}, \tag{11}$$

$$\mathcal{E}_{2,n} := \{\mathbf{x}^n \in \mathbb{R}^{2\times n} : d_Y(y^n, \hat{y}^n(\mathbf{x}^n)) > D_2\}, \tag{12}$$

$$\mathcal{E}_n := \mathcal{E}_{1,n} \cup \mathcal{E}_{2,n}, \tag{13}$$

$$\mathcal{A}_n := \mathbb{R}^{2\times n} \setminus \mathcal{E}_n. \tag{14}$$

As in the discrete second-order criterion [9, Eq. (14)], the error probability concerns the union, not two separately constrained probabilities:

$$\begin{aligned} \varepsilon_n(D_1, D_2) &:= P_{XY}^n(\mathcal{E}_n) \\ &= 1 - P_{XY}^n(\mathcal{A}_n). \end{aligned} \tag{15}$$

We abbreviate it by ε_n when the fixed distortions are clear. Blocklength is implicit in f_i and ϕ_j; their domains determine it. All codes are deterministic after the codebooks have been selected.

Definition 1 (First-order achievable rates)**.** *A tuple $\mathbf{R} = (R_0, R_1, R_2) \in \mathbb{R}_+^3$ is first-order achievable if a sequence of the codes above satisfies*

$$\limsup_{n\to\infty} \frac{\log M_{j,n}}{n} \le R_j, \qquad j \in [0:2], \tag{16}$$

$$\lim_{n\to\infty} \varepsilon_n = 0. \tag{17}$$

The closure of these tuples is the rate region $\mathcal{R}(D_1, D_2 \mid P_{XY})$.

For $(R_1, R_2) \in \mathbb{R}_+^2$, its lower common-rate boundary is

$$\mathsf{R}_0(R_1, R_2, D_1, D_2 \mid P_{XY}) := \inf\big\{r_0 \in \mathbb{R}_+ : (r_0, R_1, R_2) \in \mathcal{R}(D_1, D_2 \mid P_{XY})\big\}. \tag{18}$$

For comparison with the classical expected-distortion formulation, let $\mathcal{R}_{\mathrm{E}}(D_1, D_2 \mid P_{XY})$ be the closure of the rate tuples satisfying (16) with

$$\limsup_{n\to\infty} \mathsf{E}_{P_{XY}^n}[d_X(X^n, \hat{X}^n)] \le D_1, \tag{19}$$

$$\limsup_{n\to\infty} \mathsf{E}_{P_{XY}^n}[d_Y(Y^n, \hat{Y}^n)] \le D_2. \tag{20}$$

Write $\mathsf{R}_{0,\mathrm{E}}$ for its minimum common rate, with the same arguments as (18). In the present Gaussian quadratic model,

$$\mathcal{R}(D_1, D_2 \mid P_{XY}) = \mathcal{R}_{\mathrm{E}}(D_1, D_2 \mid P_{XY}). \tag{21}$$

Appendix A-A proves both inclusions. An error-triggered zero reconstruction gives the expected-distortion bound with at most one extra common symbol. In the opposite direction, the covariance-shell construction is used with a fixed rate margin, followed by closure. The argument does not infer expected distortion for an unmodified code from its excess-distortion probability. The second-order criterion below remains the joint excess-distortion criterion.

B. First-Order Boundary and Second-Order Criterion

We first recall the unrestricted Gray–Wyner expression, using the auxiliary W as in [9, Theorem 1 and Eqs. (11)–(13)]. For a source distribution $\tilde{P}_{XY}$ on $\mathbb{R}^2$ and private budgets $(r_1, r_2) \in \mathbb{R}_+^2$, let $\mathcal{Q}(\tilde{P}_{XY}; r_1, r_2)$ contain the joint distributions of $(X, Y, W, \hat{X}, \hat{Y})$ with factorization $\tilde{P}_{XY} Q_{W|XY} Q_{\hat{X}|XW} Q_{\hat{Y}|YW}$. The auxiliary ranges over standard Borel alphabets. The two distortion constraints are $\mathsf{E}_Q[d_X(X, \hat{X})] \le D_1$ and $\mathsf{E}_Q[d_Y(Y, \hat{Y})] \le D_2$, and the two private constraints are $I_Q(X; \hat{X} \mid W) \le r_1$ and $I_Q(Y; \hat{Y} \mid W) \le r_2$. The expected-distortion characterization [7], [12] gives

$$\mathsf{R}_{0,\mathrm{E}}(r_1, r_2, D_1, D_2 \mid P_{XY}) = \inf_{Q \in \mathcal{Q}(P_{XY}; r_1, r_2)} I_Q(X, Y; W). \tag{22}$$

For a fixed joint distribution Q_{XW}, write $R_{X|W}(Q_{XW}, D_1)$ for the infimum of $I(X; \hat{X} \mid W)$ over reconstruction kernels meeting the first distortion constraint. Define $R_{Y|W}(Q_{YW}, D_2)$ analogously. These are the two conditional rate-distortion functions in the general expression; no Gaussian restriction has yet been imposed.

For $(\mathbf{S}, \mathbf{K}) \in (\mathbb{S}_{++}^2)^2$ with $\mathbf{K} \preceq \mathbf{S}$, consider the jointly Gaussian design $\mathbf{X} = \mathbf{W} + \mathbf{N}$, where $\mathbf{W}$ and $\mathbf{N}$ are independent with covariances $\mathbf{S} - \mathbf{K}$ and $\mathbf{K}$. The vector $\mathbf{W}$ is a Gaussian realization of the unrestricted common auxiliary. Let $Q_{\mathbf{S}} := \mathcal{N}(\mathbf{0}, \mathbf{S})$ be the source marginal of this design and $Q_{\mathbf{S},X}, Q_{\mathbf{S},Y}$ its two marginals. The residual covariance $\mathbf{K}$ describes the uncertainty in (X, Y) after $\mathbf{W}$ is given. The common and private design rates are

$$I_Q(X, Y; \mathbf{W}) = \frac{1}{2} \log \frac{|\mathbf{S}|}{|\mathbf{K}|}, \tag{23}$$

$$R_{X|\mathbf{W}}(Q_{X\mathbf{W}}, D_1) = \frac{1}{2}[\log(K_{11}/D_1)]^+, \tag{24}$$

$$R_{Y|\mathbf{W}}(Q_{Y\mathbf{W}}, D_2) = \frac{1}{2}[\log(K_{22}/D_2)]^+. \tag{25}$$

The common covariance may be singular; $\mathbf{K}$ remains positive definite. Define the feasible residual matrices and their minimum common rate by

$$\mathcal{F}(\mathbf{S}; r_1, r_2) := \big\{\mathbf{K} \in \mathbb{S}_{++}^2 : \mathbf{K} \preceq \mathbf{S},\ K_{jj} \le D_j e^{2r_j},\ j \in [1:2]\big\}, \tag{26}$$

$$\mathsf{R}_0(\mathbf{S}; r_1, r_2) := \min_{\mathbf{K} \in \mathcal{F}(\mathbf{S}; r_1, r_2)} \frac{1}{2} \log \frac{|\mathbf{S}|}{|\mathbf{K}|}. \tag{27}$$

We denote this covariance-parametrized Gaussian value by R_0 to distinguish it from the operational boundary $\mathsf{R}_0(\cdot \mid P_{XY})$. Gaussian substitution only upper-bounds the expected-distortion common rate in (22). Yu's unrestricted mutual-information envelope supplies the reverse inequality [12, Theorem 3 and Corollary 2]; (21) then identifies the operational criterion used

here. Thus $\mathsf{R}_0(\mathbf{S}; r_1, r_2) = \mathsf{R}_{0,\mathrm{E}}(r_1, r_2, D_1, D_2 \mid Q_{\mathbf{S}}) = \mathsf{R}_0(r_1, r_2, D_1, D_2 \mid Q_{\mathbf{S}})$. Appendix A-A gives both arguments and the attaining reconstruction kernels. Lemma 3 realizes the covariance design for a deterministic shell; its proof does not require that operational identification.

Let $\mathbf{S} = [s_{jk}]_{j,k\in[1:2]} \in \mathbb{S}^2_{++}$, write $s_j := s_{jj}$, and restrict attention to a neighborhood of $\mathbf{\Sigma}$ where $s_j > D_j$ and $s_{12} > 0$. Define

$$\rho(\mathbf{S}) := \frac{s_{12}}{\sqrt{s_1 s_2}}, \tag{28}$$

$$R_X(Q_{\mathbf{S},X}, D_1) := \frac{1}{2}\log\frac{s_1}{D_1}, \tag{29}$$

$$R_Y(Q_{\mathbf{S},Y}, D_2) := \frac{1}{2}\log\frac{s_2}{D_2}, \tag{30}$$

$$\alpha(\mathbf{S}; r_1) := \big[R_X(Q_{\mathbf{S},X}, D_1) - r_1\big]^+, \tag{31}$$

$$\beta(\mathbf{S}; r_2) := \big[R_Y(Q_{\mathbf{S},Y}, D_2) - r_2\big]^+. \tag{32}$$

The quantities α and β are the marginal rate deficits left after the private descriptions have been allocated.

For fixed $\rho \in (0,1)$ and $(\alpha, \beta) \in \mathbb{R}^2_+$, put

$$a_\alpha := \sqrt{1 - e^{-2\alpha}}, \tag{33}$$

$$b_\beta := \sqrt{1 - e^{-2\beta}}, \tag{34}$$

$$r_{\alpha,\beta} := (\rho - a_\alpha b_\beta)e^{\alpha+\beta}. \tag{35}$$

For $(\alpha, \beta) \in \mathbb{R}^2_+$ with $\min\{\alpha, \beta\} > 0$, the four branches are indexed by

$$\mathcal{D}_1 := \left\{(\alpha,\beta) \in (0,\infty)^2 : a_\alpha b_\beta \le \rho \le \min\{b_\beta/a_\alpha, a_\alpha/b_\beta\}\right\}, \tag{36}$$

$$\mathcal{D}_2 := \left\{(\alpha,\beta) \in (0,\infty)^2 : \rho \le a_\alpha b_\beta\right\}, \tag{37}$$

$$\mathcal{D}_3 := \left\{(\alpha,\beta) \in (0,\infty)^2 : \rho > b_\beta/a_\alpha\right\}, \tag{38}$$

$$\mathcal{D}_4 := \left\{(\alpha,\beta) \in (0,\infty)^2 : \rho > a_\alpha/b_\beta\right\}. \tag{39}$$

Since $a_\alpha, b_\beta < 1$, the condition in (37) implies both ratio inequalities in Yu's expression for that branch. On shared interfaces the neighboring formulas agree. With continuous extension to the axes, define

$$\Upsilon_\rho(\alpha,\beta) := \begin{cases} \alpha + \beta - \dfrac{1}{2}\log\dfrac{1 - r^2_{\alpha,\beta}}{1-\rho^2}, & (\alpha,\beta) \in \mathcal{D}_1, \\ \alpha + \beta + \dfrac{1}{2}\log(1-\rho^2), & (\alpha,\beta) \in \mathcal{D}_2, \\ \alpha, & (\alpha,\beta) \in \mathcal{D}_3, \\ \beta, & (\alpha,\beta) \in \mathcal{D}_4. \end{cases} \tag{40}$$

In particular, $\Upsilon_\rho(\alpha, 0) = \alpha$, $\Upsilon_\rho(0, \beta) = \beta$, and $\Upsilon_\rho(0, 0) = 0$. The quantity $r_{\alpha,\beta}$ is the residual correlation in the first region and lies in $[0, 1)$.

Proposition 1 (Gaussian boundary, Yu [12])**.** *For every $\mathbf{S}$ in the neighborhood specified above and every $(r_1, r_2) \in \mathbb{R}^2_+$,*

$$\mathsf{R}_0(\mathbf{S}; r_1, r_2) = \Upsilon_{\rho(\mathbf{S})}\big(\alpha(\mathbf{S}; r_1), \beta(\mathbf{S}; r_2)\big). \tag{41}$$

At $\mathbf{S} = \mathbf{\Sigma}$, these marginal values are $R_X(P_X, D_1)$ and $R_Y(P_Y, D_2)$, respectively, and

$$\mathsf{R}_0(R_1, R_2, D_1, D_2 \mid P_{XY}) = \Upsilon_\rho\big([R_X(P_X, D_1) - R_1]^+, [R_Y(P_Y, D_2) - R_2]^+\big). \tag{42}$$

Appendix A-A derives this existing result from the general expression and identifies the Gaussian attaining design.

Proposition 1 determines the whole first-order region: a nonnegative rate tuple is feasible precisely when its common rate is at least the right-hand side of (42). Thus fixing the two private rates determines the least common rate. The following definition selects the Pareto-minimal boundary tuples without a redundant description rate.

Definition 2 (Pareto-Minimal Boundary Rate Tuple)**.** *A tuple $\mathbf{R}^* = (R_0^*, R_1^*, R_2^*) \in \mathcal{R}(D_1, D_2 \mid P_{XY})$ is a Pareto-minimal boundary rate tuple if reducing any positive coordinate, with the other two fixed, leaves the rate region. Let $\mathbf{e}_j$ denote the coordinate vector of $\mathbb{R}^3$ indexed by $j \in [0:2]$. Equivalently,*

$$\mathbf{R}^* - t\mathbf{e}_j \notin \mathcal{R}(D_1, D_2 \mid P_{XY}), \qquad j \in [0:2], \quad 0 < t \le R_j^*. \tag{43}$$

The qualification in this definition matters only when a coordinate is redundant; a point on the lower common-rate graph alone need not satisfy it. Every such Pareto-minimal boundary tuple has $0 \le R_j^* \le \frac{1}{2}\log(\Sigma_{jj}/D_j)$ for $j \in [1:2]$. Its deficits are

$$\alpha^* := R_X(P_X, D_1) - R_1^*, \tag{44}$$

$$\beta^* := R_Y(P_Y, D_2) - R_2^*. \tag{45}$$

For (α, β) in the rectangle $\mathcal{I}_{\mathbf{\Sigma}} := [0, R_X(P_X, D_1)] \times [0, R_Y(P_Y, D_2)]$, the associated lower-boundary tuple is

$$\mathbf{R}(\alpha, \beta) := \big(\Upsilon_\rho(\alpha, \beta), R_X(P_X, D_1) - \alpha, R_Y(P_Y, D_2) - \beta\big). \tag{46}$$

This gives a direct rule for finding the boundary: compute the deficits, select one of the four regions in (36)–(39), and evaluate (40). The four geometric cases are the coupled curved part $\mathcal{D}_1$, the sum-rate part $\mathcal{D}_2$, and the two one-source parts $\mathcal{D}_3, \mathcal{D}_4$. On the latter three,

$$R_0 + R_1 + R_2 = R_X(P_X, D_1) + R_Y(P_Y, D_2) + \tfrac{1}{2}\log(1-\rho^2), \qquad (\alpha, \beta) \in \mathcal{D}_2, \tag{47}$$

$$R_0 + R_1 = R_X(P_X, D_1), \qquad (\alpha, \beta) \in \mathcal{D}_3, \tag{48}$$

$$R_0 + R_2 = R_Y(P_Y, D_2), \qquad (\alpha, \beta) \in \mathcal{D}_4. \tag{49}$$

Within $\mathcal{I}_{\mathbf{\Sigma}}$, every lower-boundary tuple $\mathbf{R}(\alpha, \beta)$ with $\alpha, \beta > 0$ and $(\alpha, \beta) \in \mathcal{D}_1 \cup \mathcal{D}_2$ is Pareto-minimal, including the finite interfaces with the one-source branches. A tuple in $\mathcal{D}_3$ is Pareto-minimal if and only if $R_2 = 0$, and a tuple in $\mathcal{D}_4$ is Pareto-minimal if and only if $R_1 = 0$. On either deficit axis away from the origin, the private rate corresponding to the vanishing deficit is positive and can be reduced slightly without changing the common rate; these tuples are therefore not Pareto-minimal. The deficit origin is Pareto-minimal and is treated below. These cases exhaust $\mathcal{I}_{\mathbf{\Sigma}}$ and include tuples with one or both private rates equal to zero. On the curved and sum-rate interiors, increasing either deficit strictly increases the common rate. At a finite interface with a one-source branch, increasing the deficit in the zero-slope coordinate enters the curved branch and produces the strict quadratic increase in (222); the other interface is symmetric. Thus a zero slope at such an interface does not make a positive private rate redundant.

There are two cases for the second-order analysis, regardless of which of the four formulas selects the first-order value. Every Pareto-minimal boundary tuple except

$$\mathbf{R}_{\mathrm{c}}^* := \big(0, R_X(P_X, D_1), R_Y(P_Y, D_2)\big) \tag{50}$$

has $R_0^* > 0$. The tuple (50) is the unique Pareto-minimal boundary tuple with zero common rate: (40) has value zero only when both deficits vanish.

For comparison, the endpoint source models described in Section II-A have the first-order requirements $R_0 \ge \alpha + \beta$ at $\rho = 0$ and $R_0 \ge \max\{\alpha, \beta\}$ at $|\rho| = 1$, with the same definitions of the marginal deficits. Independence makes the common budget split into two separate source descriptions; perfect correlation turns it into a description of the same scalar Gaussian source for both receivers. Appendix A-A justifies these reductions. They are not an extension of the second-order theorem to those endpoints.

Definition 3 (Second-order achievable offsets)**.** *Fix a Pareto-minimal boundary tuple $\mathbf{R}^*$ from Definition 2 and $\varepsilon \in (0,1)$. A vector $\mathbf{L} = (L_0, L_1, L_2) \in \mathbb{R}^3$ is second-order $(R_0^*, R_1^*, R_2^*, D_1, D_2, \varepsilon)$-achievable if a code sequence satisfies*

$$\limsup_{n\to\infty} \frac{\log M_{j,n} - nR_j^*}{\sqrt{n}} \le L_j, \qquad j \in [0:2], \tag{51}$$

$$\limsup_{n\to\infty} \varepsilon_n \le \varepsilon. \tag{52}$$

The closure of these vectors is denoted by $\mathcal{L}(R_0^*, R_1^*, R_2^*, D_1, D_2, \varepsilon)$, following [9, Definition 3].

We use $\mathcal{L}(\mathbf{R}^*, D_1, D_2, \varepsilon)$ as its tuple abbreviation; the source distribution is fixed throughout.

Define the zero-rate index set

$$\mathcal{Z}(\mathbf{R}^*) := \{j \in [0:2] : R_j^* = 0\}. \tag{53}$$

Since message alphabets are nonempty, the offsets must satisfy $L_j \ge 0$ for $j \in \mathcal{Z}(\mathbf{R}^*)$. Importantly, $R_0^* = 0$ permits a common description of order $\sqrt{n}$; it does not impose $M_{0,n} = 1$.

III. Main Results and Discussions

We state the exact region, explain its scalar and joint fluctuation mechanisms, and then evaluate representative operating points.

A. Exact Second-Order Region

The theorem has two cases. At positive common rate, the supporting slopes determine a scalar dispersion. At zero common rate, a nonlinear allocation function acts on the two marginal-power fluctuations.

Proposition 2 (Regularity at positive common rate)**.** *For every Pareto-minimal boundary tuple $\mathbf{R}^*$ of Definition 2 with $R_0^* > 0$, the function $(\mathbf{S}, r_1, r_2) \mapsto \mathsf{R}_0(\mathbf{S}; r_1, r_2)$ has a continuously differentiable extension to an open neighborhood of $(\mathbf{\Sigma}, R_1^*, R_2^*)$.*

Appendix A-A derives this property from Yu's formula, including the finite branch interfaces and zero private rates. The extension defines derivatives at zero private rates; operational rates remain nonnegative. Thus no additional differentiability assumption is required within the source model and boundary class considered here.

Fix a Pareto-minimal boundary tuple with $R_0^* > 0$ and deficits (α^*, β^*) from (44)–(45). Define the supporting slopes by

$$\lambda_1^* := -\left.\frac{\partial}{\partial r_1}\mathsf{R}_0(r_1, R_2^*, D_1, D_2 \mid P_{XY})\right|_{r_1 = R_1^*} = \partial_\alpha \Upsilon_\rho(\alpha^*, \beta^*), \tag{54}$$

$$\lambda_2^* := -\left.\frac{\partial}{\partial r_2}\mathsf{R}_0(R_1^*, r_2, D_1, D_2 \mid P_{XY})\right|_{r_2 = R_2^*} = \partial_\beta \Upsilon_\rho(\alpha^*, \beta^*). \tag{55}$$

Write $\boldsymbol{\lambda}^* := (\lambda_1^*, \lambda_2^*) \in [0,1]^2$. As in [9, Eq. (22)], $(1, \lambda_1^*, \lambda_2^*)$ is the supporting normal in rate coordinates. Define

$$V(R_1^*, R_2^*, D_1, D_2 \mid P_{XY}) := \frac{1 + (\lambda_1^* + \lambda_2^* - 1)^2}{2}. \tag{56}$$

We abbreviate this quantity by V. Appendix A-B proves its variance interpretation and the bounds

$$1 \le \lambda_1^* + \lambda_2^* \le 2, \tag{57}$$

$$\frac{1}{2} \le V \le 1. \tag{58}$$

At $\mathbf{R}_{\mathrm{c}}^*$, a common description must supply the pair of residual private-rate deficits jointly. We define the function describing this allocation through an optimization, rather than through an ordinary derivative. For fixed $\rho \in (0,1)$, let

$$\mathcal{H}_\rho := \big\{(p,q) \in \mathbb{R}_+^2 : \min\{p,q\} > 1,\ (p-1)(q-1) \ge \rho^2\big\}. \tag{59}$$

For $(a,b) \in \mathbb{R}_+^2$, define

$$\psi_\rho(a,b) := \sup_{(p,q)\in\mathcal{H}_\rho} \left(\frac{a}{p} + \frac{b}{q}\right). \tag{60}$$

The feasible exponents are those in the Gaussian hypercontractive inequality used in Section V-B. The supremum includes the limits $(p,q) \to (1,\infty)$ and $(\infty,1)$, not finite exponents equal to one. Solving (60) gives

$$\psi_\rho(a,b) = \begin{cases} \dfrac{a + b - 2\rho\sqrt{ab}}{1-\rho^2}, & \rho^2 a \le b \le a/\rho^2, \\ a, & b < \rho^2 a, \\ b, & a < \rho^2 b. \end{cases} \tag{61}$$

The value at $(0,0)$ is zero. Appendix B derives this closed form from the single definition (60).

Let

$$\mathbf{Z} = (Z_1, Z_2)^{\mathsf{T}} \sim \mathcal{N}\left(\mathbf{0}, \frac{1}{2}\begin{bmatrix} 1 & \rho^2 \\ \rho^2 & 1 \end{bmatrix}\right), \tag{62}$$

and denote its distribution by $P_{\mathbf{Z}}$. The two coordinates are the limiting marginal-power fluctuations. The common budget is applied to their two positive private deficits through (60).

The source distribution, distortions, and Pareto-minimal boundary tuple in the following theorem are fixed as $n \to \infty$.

Theorem 1. *For the source $P_{XY} = \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$ and distortions of Section II-A, every Pareto-minimal boundary tuple $\mathbf{R}^* = (R_0^*, R_1^*, R_2^*)$ of Definition 2, and every $\varepsilon \in (0,1)$, the region $\mathcal{L}(\mathbf{R}^*, D_1, D_2, \varepsilon)$ is as follows. At nonzero common rate, λ_1^*, λ_2^* are defined by* (54)–(55) *and V by* (56)*; at zero common rate, $P_{\mathbf{Z}}$ and ψ_ρ are defined by* (62) *and* (60)*.*

(i) At a point with nonzero common rate $R_0^* > 0$,

$$\mathcal{L}(\mathbf{R}^*, D_1, D_2, \varepsilon) = \Big\{\mathbf{L} \in \mathbb{R}^3 : L_0 + \lambda_1^* L_1 + \lambda_2^* L_2 \ge \sqrt{V}\,\mathrm{Q}^{-1}(\varepsilon), \quad L_j \ge 0\ \forall j \in \mathcal{Z}(\mathbf{R}^*)\Big\}. \tag{63}$$

(ii) At the zero common rate point $\mathbf{R}^* = \mathbf{R}_{\mathrm{c}}^*$,

$$\mathcal{L}(\mathbf{R}_{\mathrm{c}}^*, D_1, D_2, \varepsilon) = \Big\{\mathbf{L} \in \mathbb{R}^3 : L_0 \ge 0, \quad P_{\mathbf{Z}}\big\{\psi_\rho([Z_1 - L_1]^+, [Z_2 - L_2]^+) \le L_0\big\} \ge 1 - \varepsilon\Big\}. \tag{64}$$

Proof sketch. The encoder groups blocks by empirical covariance, selects the Gaussian design for each group, and combines a common covering with two private coverings. The error is bounded by the event that the local first-order common rate exceeds the available budget, apart from an atypical-covariance probability. At nonzero common rate, an ordinary boundary expansion yields an independent sum of centered quadratic information quantities and a scalar Gaussian limit; at zero common rate, a directional expansion retains the two private fluctuations and gives a bivariate limit. The converse uses the specialized positive-weight Gaussian bound in Lemma 7, a scalar bound for one-source directions, and Lemma 8 at the zero common rate corner. The direct and converse limits agree. Sections IV and V give the two proofs. □

B. Boundary Fluctuations and Interpretation

We explain the fluctuation behind the scalar result and the joint allocation mechanism at the corner. The explicit branch derivatives used below are collected in Appendix A-A.

Fix a Pareto-minimal boundary tuple $\mathbf{R}^*$ with $R_0^* > 0$, and recall the source covariance $\mathbf{\Sigma}$. For $(x, y) \in \mathbb{R}^2$, set $\mathbf{x} := (x, y)^{\mathsf{T}}$. Define $\mathbf{G} \in \mathbb{S}^2$ and the Gaussian covariance influence function by

$$\mathbf{G} := \nabla_{\mathbf{S}} \mathsf{R}_0(\mathbf{S}; R_1^*, R_2^*)|_{\mathbf{S}=\mathbf{\Sigma}}\,, \tag{65}$$

$$\jmath_{XY}(x, y \mid R_1^*, R_2^*, D_1, D_2, P_{XY}) := R_0^* + \mathbf{x}^{\mathsf{T}}\mathbf{G}\mathbf{x} - \operatorname{tr}(\mathbf{G}\mathbf{\Sigma}). \tag{66}$$

For the fixed operating point, abbreviate this function by $\jmath_{XY}(x, y)$. Its argument order follows [9, Eq. (26)], but the definition here is explicitly the rate-normalized Gaussian covariance influence function.

The chosen additive constant gives the normalization

$$\mathsf{E}_{P_{XY}}[\jmath_{XY}(X, Y)] = R_0^*. \tag{67}$$

Moreover, for $Q_{\mathbf{S}} := \mathcal{N}(\mathbf{0}, \mathbf{S})$ and $\mathbf{S} \to \mathbf{\Sigma}$,

$$\mathsf{R}_0(\mathbf{S}; R_1^*, R_2^*) = \mathsf{E}_{Q_{\mathbf{S}}}[\jmath_{XY}(X, Y)] + o(\|\mathbf{S} - \mathbf{\Sigma}\|). \tag{68}$$

The centered first variation in (68) is the quantity used in both coding directions. Appendix A-B proves these identities and evaluates $\mathbf{G}$ explicitly; no separate uncentered tilted-density identity is needed.

The dispersion in (56) also has the matrix representation

$$\begin{aligned} V(R_1^*, R_2^*, D_1, D_2 \mid P_{XY}) &= \mathsf{Var}_{P_{XY}}[\jmath_{XY}(X, Y)] \\ &= 2\operatorname{tr}[(\mathbf{G}\mathbf{\Sigma})^2]. \end{aligned} \tag{69}$$

The argument order agrees with [9, Eq. (30)].

On the affine branches and their adjoining interfaces,

$$V = \begin{cases} 1, & \text{if } (\alpha^*, \beta^*) \in \mathcal{D}_2, \\ \frac{1}{2}, & \text{if } (\alpha^*, \beta^*) \in \mathcal{D}_3 \cup \mathcal{D}_4. \end{cases} \tag{70}$$

A zero slope need not make a private description redundant. At a $\mathcal{D}_1$–$\mathcal{D}_3$ interface with $R_2^* > 0$, one has $b_{\beta^*} = \rho a_{\alpha^*}$ and $(\lambda_1^*, \lambda_2^*) = (1, 0)$. Reducing R_2^* by a small amount enters the curved region and increases the required common rate quadratically, as shown in (222). Thus the coordinate remains necessary for Pareto minimality. A private-rate perturbation of order $n^{-1/2}$ affects the common rate by only $O(n^{-1})$ in this direction, which explains the absence of L_2 from (63). The other one-source interface is symmetric.

Remark 1 (Spectrum and slope-only dispersion)**.** *At $R_0^* > 0$, the eigenvalues of the symmetric matrix $\mathbf{\Sigma}^{1/2}\mathbf{G}\mathbf{\Sigma}^{1/2}$ are $1/2$ and $(\lambda_1^* + \lambda_2^* - 1)/2$. Consequently, the dispersion depends on the supporting slopes only through their sum, as in* (56)*. This does not reduce the whole region to that sum: its rate displacement remains $L_0 + \lambda_1^* L_1 + \lambda_2^* L_2$. Nor does it imply correlation independence at a fixed rate tuple, because the slopes themselves depend on the source and the selected tuple. The matrix $\mathbf{G}$ is still needed to identify the empirical fluctuation and match the converse. Appendix A-B proves the eigenvalue statement, including affine interfaces and zero private rates.*

There is no path-independent scalar dispersion at the corner. For sufficiently small $t > 0$, use the two nonredundant tuples from (46) *with deficit pairs (t, t) and (t, β_t), where*

$$\beta_t := -\tfrac{1}{2}\log(1 - \rho^2 + \rho^2 e^{-2t}). \tag{71}$$

Along (t, t), symmetry and $r_{t,t} = 1 - (1-\rho)e^{2t}$ give both slopes equal to $1/(1 + r_{t,t})$. Along the second path, $b_{\beta_t} = \rho a_t$ places the tuple on the $\mathcal{D}_1$–$\mathcal{D}_3$ interface, with slopes $(1, 0)$. If $V^{(1)}(t), V^{(2)}(t)$ denote their dispersions, then

$$\lim_{t\downarrow 0} V^{(1)}(t) = \frac{1}{2}\left[1 + \left(\frac{2}{1+\rho} - 1\right)^2\right] = \frac{1+\rho^2}{(1+\rho)^2}, \tag{72}$$

$$\lim_{t\downarrow 0} V^{(2)}(t) = \frac{1}{2}. \tag{73}$$

where Eqs. (72)–(73) *substitute the respective slope limits into* (56)*. Both paths converge to $\mathbf{R}_{\mathrm{c}}^*$, but the limits differ for $0 < \rho < 1$. Theorem 1 fixes its operating tuple before the blocklength limit; it does not assert a uniform transition theorem for an n-dependent tuple approaching the corner.*

Remark 2 (Parallel with the discrete lossy theorem)**.** *Under the differentiability and support conditions of [9, Theorem 4], the discrete result has the same weighted-half-space form as* (63)*. To distinguish the two operating points in this comparison only, attach a subscript* D *to its slopes and dispersion:*

$$L_0 + \lambda_{1,\mathrm{D}}^* L_1 + \lambda_{2,\mathrm{D}}^* L_2 \geq \sqrt{V_{\mathrm{D}}}\, \mathrm{Q}^{-1}(\varepsilon). \tag{74}$$

The definition of each slope is the same negative private-rate derivative; the values depend on the source. The discrete source derivative is represented by centered Gray–Wyner tilted information. Here, the covariance derivative satisfies

$$\sqrt{n}\left\langle \mathbf{G}, \hat{\mathbf{\Sigma}}_n - \mathbf{\Sigma} \right\rangle = \frac{1}{\sqrt{n}} \sum_{i\in[1:n]} [\jmath_{XY}(X_i, Y_i) - R_0^*]. \tag{75}$$

Lemma 3 replaces the generalized type-covering lemma in [9, Lemma 8]. Local expansion and a central limit theorem then complete the direct argument in the same order. Table I compares the centered quantities; the Gaussian auxiliary and empirical descriptor retain the domains specified in Section II-A. At the corner, the directional expansion and Lemma 8 replace the scalar reduction.

TABLE I
THE LOCAL-BOUNDARY REDUCTION AT DIFFERENTIABLE POINTS. THE DISCRETE COLUMN REFERS TO [9, THEOREM 4 AND SECTION IV-A].

Role	Discrete lossy Gray–Wyner	Gaussian lossy Gray–Wyner
Empirical descriptor	Joint type $\hat{T}_{x^n y^n}$	Empirical matrix $\hat{\mathbf{S}}(\mathbf{x}^n)$
Local benchmark	Source-distribution-dependent common rate	Covariance-dependent common rate from Yu's formula
Uniform coding result	Generalized type covering	Covariance-shell covering, Lemma 3
Centered first variation	Centered tilted information	$\mathbf{x}^{\mathsf{T}}\mathbf{G}\mathbf{x} - \mathrm{tr}(\mathbf{G}\mathbf{\Sigma})$
Dispersion	Variance of that first variation	$[1 + (\lambda_1^* + \lambda_2^* - 1)^2]/2$
Rate offset	$L_0 + \lambda_{1,\mathrm{D}}^* L_1 + \lambda_{2,\mathrm{D}}^* L_2$	$L_0 + \lambda_1^* L_1 + \lambda_2^* L_2$

At the corner, Lemma 5 establishes the uniform directional expansion underlying

$$\lim_{t\downarrow 0} \frac{\Upsilon_\rho(ta, tb)}{t} = \psi_\rho(a, b), \qquad (a, b) \in \mathbb{R}_+^2. \tag{76}$$

Appendix B treats the moving interfaces and covariance perturbations in detail. The values $\psi_\rho(a, 0) = a$ and $\psi_\rho(a, a) = 2a/(1+\rho)$ show how a common description serves balanced and unbalanced deficits differently. The correlation ρ enters this allocation cost, whereas ρ^2 is the correlation of the limiting squared-source fluctuations in (62). Consequently, the corner offset region depends only on ρ and ε; the marginal variances and distortions determine its first-order location.

Remark 3 (Scope of the existing converse and the zero common rate case)**.** *The positive-weight converse is imported from [11, Theorem 17]; Lemma 7 states its specialization after Lemma 6 identifies the support and fluctuation. A scalar Gaussian converse handles a zero weight. At the corner, separate fixed-direction bounds cannot be combined by moving a supremum inside a probability. The decisive additional statement is the finite-blocklength conditional-success bound* (191) *in Lemma 8: it holds uniformly over codes and covariance cells, with the supporting-direction optimization already inside its deterministic exponent. Applying that bound before the bivariate limit preserves the joint success event.*

Remark 4 (Zero common message and joint success)**.** *At $\mathbf{R}_{\mathrm{c}}^*$ and $L_0 = 0$, the condition in* (64) *is*

$$P_{\mathbf{Z}}\{Z_1 \le L_1, Z_2 \le L_2\} = \Phi_2(\sqrt{2}L_1, \sqrt{2}L_2; \rho^2) \ge 1 - \varepsilon, \tag{77}$$

where $\Phi_2(\cdot, \cdot; r)$ is the standard bivariate Gaussian CDF with correlation $r \in (-1, 1)$. Indeed,

$$\begin{aligned} &P_{\mathbf{Z}}\{\psi_\rho([Z_1 - L_1]^+, [Z_2 - L_2]^+) \le 0\} \\ &\quad = P_{\mathbf{Z}}\{[Z_1 - L_1]^+ = [Z_2 - L_2]^+ = 0\} && (78)\\ &\quad = P_{\mathbf{Z}}\{Z_1 \le L_1, Z_2 \le L_2\} && (79)\\ &\quad = \Phi_2(\sqrt{2}L_1, \sqrt{2}L_2; \rho^2), && (80)\end{aligned}$$

where Eq. (78) *uses the zero set of ψ_ρ, Eq.* (79) *uses the positive parts, and Eq.* (80) *rescales* (62)*.*

In fact, an absent common message attains this entire slice. Let $\mathcal{L}_\varnothing(D_1, D_2, \varepsilon)$ be the closure of the private offset pairs satisfying Definition 3 at the two marginal first-order rates with $M_{0,n} = 1$ for every n. Then

$$\mathcal{L}_\varnothing(D_1, D_2, \varepsilon) = \{(L_1, L_2) \in \mathbb{R}^2 : (0, L_1, L_2) \in \mathcal{L}(\mathbf{R}_{\mathrm{c}}^*, D_1, D_2, \varepsilon)\}. \tag{81}$$

The private-ball construction in Section IV-A and its analysis in Section IV-C prove achievability without a common header. The converse follows from the arbitrary-code bound in Section V-B. Thus the $o(\sqrt{n})$ common nats allowed by $L_0 = 0$ are unnecessary for this second-order slice.

The remaining coupling is probabilistic, not a need for common communication. For $\ell_\varepsilon := \mathrm{Q}^{-1}(\varepsilon)/\sqrt{2}$,

$$\begin{aligned} P_{\mathbf{Z}}\{Z_1 \le \ell_\varepsilon, Z_2 \le \ell_\varepsilon\} &= P_{\mathbf{Z}}\{Z_1 \le \ell_\varepsilon\} - P_{\mathbf{Z}}\{Z_1 \le \ell_\varepsilon, Z_2 > \ell_\varepsilon\} && (82)\\ &< 1 - \varepsilon, && (83)\end{aligned}$$

where Eq. (82) *partitions the marginal event, and Eq.* (83) *uses the strictly positive joint Gaussian density. Each marginal error may equal ε while the joint target fails. A genuinely second-order common description can improve the private offsets only when $L_0 > 0$.*

C. Examples and Numerical Evaluation

For a symmetric nonzero common rate tuple, set $\sigma_X^2 = \sigma_Y^2 = 1$, $D_1 = D_2 = D$, and $\lambda_1^* = \lambda_2^* = \lambda$. The source correlation is $\rho \in (0, 1)$, and choose

$$\frac{1}{1+\rho} < \lambda < 1, \qquad 0 < D < \frac{\lambda(1-\rho)}{2\lambda - 1}. \tag{84}$$

The residual covariance has eigenvalues $(1-\rho)/(2\lambda-1)$ and $1-\rho$ in the sum and difference directions. The rate tuple and dispersion are

$$\begin{aligned} R_0^* &= \frac{1}{2}\log\frac{(1+\rho)(2\lambda-1)}{1-\rho}, && (85)\\ R_1^* &= \frac{1}{2}\log\frac{\lambda(1-\rho)}{(2\lambda-1)D}, && (86)\\ R_2^* &= R_1^*, && (87)\\ V &= \frac{1 + (2\lambda-1)^2}{2}. && (88)\end{aligned}$$

At $\rho = 0.6$, $D = 0.2$, and $\lambda = 0.8$, one obtains $R_0^* = 0.4377$, $R_1^* = R_2^* = 0.4904$, and $V = 0.68$. For $L_1 = L_2 = 0$ and $\varepsilon = 0.1$, the normal approximation is

$$R_{0,n}^{(2)} := R_0^* + \sqrt{V/n}\,\mathrm{Q}^{-1}(\varepsilon). \tag{89}$$

It gives 0.5434, 0.4850, and 0.4712 at $n = 100, 500, 1000$. These numbers evaluate an asymptotic expansion; the covering proof does not certify an operational finite-blocklength optimum or a numerically small remainder at these blocklengths.

We next consider unequal supporting slopes near a one-source interface. Fix $\rho = 0.6$, $\sigma_X^2 = \sigma_Y^2 = 1$, $D_1 = D_2 = 0.2$, and $\alpha^* = 0.5$. Define

$$\beta_{\mathrm{int}} := -\frac{1}{2}\log\left(1 - \rho^2 + \rho^2 e^{-2\alpha^*}\right) \simeq 0.1291026732. \tag{90}$$

The pair $(\alpha^*, \beta_{\mathrm{int}})$ lies on the $\mathcal{D}_1$–$\mathcal{D}_3$ interface. Table II evaluates three Pareto-minimal tuples with $R_1^* = R_X(P_X, D_1) - \alpha^* \simeq 0.3047189562$; the first two lie in the curved region. For the middle row, the supporting displacement in (63) is $L_0 + 0.982442L_1 + 0.103022L_2$. Thus equal increments in the two private offsets have different effects. At the interface, the condition reduces to $L_0 + L_1 \ge \mathrm{Q}^{-1}(\varepsilon)/\sqrt{2}$. The second private rate remains nonredundant: reducing it increases β^* into the

TABLE II
ASYMMETRIC BOUNDARY TUPLES NEAR THE $\mathcal{D}_1$–$\mathcal{D}_3$ INTERFACE. THE FIXED PARAMETERS ARE $\rho = 0.6$, UNIT MARGINAL VARIANCES, $D_1 = D_2 = 0.2$, $\alpha^* = 0.5$, AND $R_1^* \simeq 0.3047189562$. RATES ARE IN NATS PER SOURCE PAIR.

$\beta^* - \beta_{\mathrm{int}}$	R_0^*	R_2^*	λ_1^*	λ_2^*	V
0.05	0.511027	0.625616	0.935913	0.396356	0.555201
0.01	0.500528	0.665616	0.982442	0.103022	0.503652
0	0.500000	0.675616	1.000000	0.000000	0.500000

curved region and raises the required common rate. Its zero supporting slope means that this increase is only quadratic in a sufficiently small private-rate perturbation, and hence does not contribute on the second-order scale.

At the zero common rate corner, fix $\rho, \varepsilon \in (0,1)$. The least symmetric private offset is

$$\ell_\rho^*(L_0,\varepsilon) := \inf\left\{\ell \in \mathbb{R} : P_{\mathbf{Z}}\left\{\psi_\rho([Z_1-\ell]^+,[Z_2-\ell]^+) \le L_0\right\} \ge 1-\varepsilon\right\}, \qquad L_0 \ge 0. \tag{91}$$

For $0 \le a \le L_0$, the largest b satisfying $\psi_\rho(a,b) \le L_0$ is

$$b_{\rho,L_0}(a) := \begin{cases} L_0, & a \in [0,\rho^2 L_0], \\ \left[\rho\sqrt{a} + \sqrt{(1-\rho^2)(L_0-a)}\right]^2, & a \in (\rho^2 L_0, L_0]. \end{cases} \tag{92}$$

For $a > L_0$ no nonnegative b is feasible. At $L_0 = 0$, the definition gives $b_{\rho,0}(0) = 0$. Conditioning the Gaussian vector in (62) on its first coordinate yields

$$\begin{aligned} &P_{\mathbf{Z}}\left\{\psi_\rho([Z_1-\ell]^+,[Z_2-\ell]^+) \le L_0\right\} \\ &\quad = \int_{-\infty}^{\ell+L_0} \frac{e^{-z^2}}{\sqrt{\pi}}\, \Phi\left(\frac{\ell + b_{\rho,L_0}([z-\ell]^+) - \rho^2 z}{\sqrt{(1-\rho^4)/2}}\right) dz, \end{aligned} \tag{93}$$

where Eq. (93) uses marginal variance $1/2$, conditional mean $\rho^2 z$, conditional variance $(1-\rho^4)/2$, and the section (92). The probability is continuous and strictly increasing in ℓ, with limits zero and one. Thus a bracketed scalar root search determines (91).

To separate fixed splitting from a sum-of-deficits allocation, define three success sets for $\ell \in \mathbb{R}$ and $L_0 \ge 0$:

$$\mathcal{A}_{\mathrm{e}}(\ell,L_0) := \{\mathbf{z} \in \mathbb{R}^2 : \max\{z_1,z_2\} \le \ell + L_0/2\}, \tag{94}$$

$$\mathcal{A}_{\mathrm{s}}(\ell,L_0) := \{\mathbf{z} \in \mathbb{R}^2 : [z_1-\ell]^+ + [z_2-\ell]^+ \le L_0\}, \tag{95}$$

$$\mathcal{A}_{\mathrm{j}}(\ell,L_0) := \{\mathbf{z} \in \mathbb{R}^2 : \psi_\rho([z_1-\ell]^+,[z_2-\ell]^+) \le L_0\}. \tag{96}$$

The elementary bound $\psi_\rho(a,b) \le a+b$ gives

$$\mathcal{A}_{\mathrm{e}}(\ell,L_0) \subseteq \mathcal{A}_{\mathrm{s}}(\ell,L_0) \subseteq \mathcal{A}_{\mathrm{j}}(\ell,L_0). \tag{97}$$

The events charge a fixed equal split, an adaptive split at sum cost, and the joint cost, respectively. Their sufficient offset benchmarks are

$$\ell_{\rho,\mathrm{sum}}(L_0,\varepsilon) := \inf\{\ell \in \mathbb{R} : P_{\mathbf{Z}}(\mathcal{A}_{\mathrm{s}}(\ell,L_0)) \ge 1-\varepsilon\}, \tag{98}$$

$$\ell_{\rho,\mathrm{split}}(L_0,\varepsilon) := \ell_\rho^*(0,\varepsilon) - L_0/2. \tag{99}$$

Consequently,

$$\ell_\rho^*(L_0,\varepsilon) \le \ell_{\rho,\mathrm{sum}}(L_0,\varepsilon) \le \ell_{\rho,\mathrm{split}}(L_0,\varepsilon). \tag{100}$$

where Eq. (100) follows from the inclusions at the same joint error target. The benchmarks are sufficient conditions, not optima of restricted coding architectures.

For the sum benchmark, conditional Gaussian integration gives

$$P_{\mathbf{Z}}(\mathcal{A}_{\mathrm{s}}(\ell,L_0)) = \int_{-\infty}^{\ell+L_0} \frac{e^{-z^2}}{\sqrt{\pi}} \Phi\left(\frac{\ell + L_0 - [z-\ell]^+ - \rho^2 z}{\sqrt{(1-\rho^4)/2}}\right) dz. \tag{101}$$

Scalar root searches evaluate the three curves in Fig. 2; they coincide at $L_0 = 0$. For $\rho = 0.6$, $\varepsilon = 0.1$, and $L_0 = 1$, their offsets are 0.1582, 0.3012, and 0.6309. The intermediate curve separates adaptive allocation from reduction of the common cost through correlation.

Table III illustrates the two roles of correlation at the corner. When $L_0 = 0$, only the correlation ρ^2 of the marginal-power fluctuations enters the joint success condition. For $L_0 = 1$, the correlation ρ also enters the common allocation cost ψ_ρ. On the

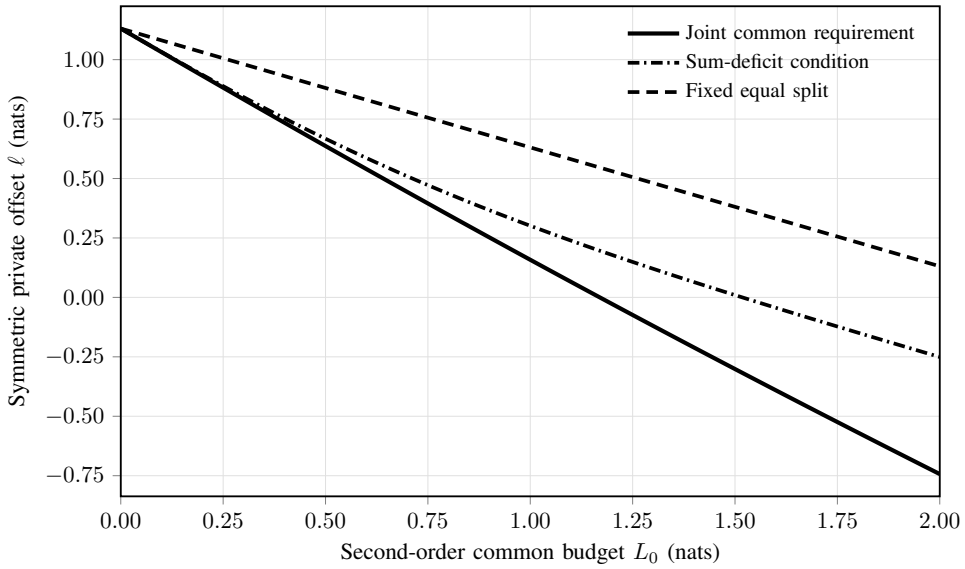


Fig. 2. Symmetric private offsets at the zero first-order common-rate tuple for $\rho = 0.6$, $\varepsilon = 0.1$, unit marginal variances, and $D_1 = D_2 = 0.2$. The curves are the exact joint condition (91), the sufficient sum-deficit condition (98), and the sufficient fixed equal-split condition (99). All use the same joint success probability; no finite-blocklength code or restricted-architecture optimum is evaluated.

TABLE III
CORRELATION SENSITIVITY OF THE LEAST SYMMETRIC PRIVATE OFFSET AT THE ZERO FIRST-ORDER COMMON-RATE TUPLE, WITH $\varepsilon = 0.1$. OFFSETS ARE IN NATS PER SOURCE PAIR.

ρ	$\ell^*_\rho(0, 0.1)$	$\ell^*_\rho(1, 0.1)$
0.2	1.152489	0.225108
0.6	1.130872	0.158202
0.9	1.052717	0.057114

displayed correlation grid, the least symmetric private offset decreases for each budget. The common budget of one nat on the second-order scale reduces the private offset by 0.927381, 0.972670, and 0.995604, respectively. These values evaluate the limiting offset region, with independent quadrature checks supplied in the numerical code.

At a fixed nonzero common rate tuple, set $\nu := \lambda_1^* + \lambda_2^* - 1 \in [0, 1]$ and define the linearized source statistic

$$\mathsf{A}_n := \sqrt{n} \left\langle \mathbf{G}, \hat{\mathbf{\Sigma}}_n - \mathbf{\Sigma} \right\rangle, \tag{102}$$

$$\mathsf{A}_n \overset{d}{=} \frac{\Xi_{1,n} - n}{2\sqrt{n}} + \frac{\nu}{2\sqrt{n}} (\Xi_{2,n} - n), \tag{103}$$

where $\Xi_{1,n}, \Xi_{2,n}$ are independent chi-squared variables with n degrees of freedom. Appendix A-B proves this exact distributional identity. Let H_n be the CDF of A_n, and let $f_{\chi^2_n}, F_{\chi^2_n}$ be the chi-squared density and CDF, with the latter set to zero at negative arguments. Then

$$H_n(x) = \int_0^\infty f_{\chi^2_n}(u) F_{\chi^2_n}\left(n(1+\nu) + 2\sqrt{n}\, x - \nu u\right) du. \tag{104}$$

Figure 3 plots $H_n(x) - \Phi(x/\sqrt{V})$ for the symmetric tuple above, where $\nu = 0.6$. On the plotted grid, the maximum absolute CDF differences are 0.03215, 0.01440, and 0.00644 for $n = 20, 100, 500$. The corresponding 0.9 quantiles are 1.08932, 1.07390, and 1.06495, compared with the Gaussian value $\sqrt{V}\, \mathrm{Q}^{-1}(0.1) = 1.05679$.

The source package supplies a reproducible numerical script, CSV data, quadrature details, and checks against every embedded plot coordinate. The absolute and relative quadrature tolerances are 2×10^{-10}, and the root tolerance is 2×10^{-11}. The asymmetric example is checked by numerical differentiation of the boundary, and the correlation-sensitivity table is checked by a second quadrature in independent Gaussian sum and difference coordinates. These are asymptotic-formula and source-statistic evaluations, not code simulations; nonlinear Taylor remainders and covering overhead are not included.

IV. ACHIEVABILITY PROOF

We prove the direct part of Theorem 1. The empirical joint type $\hat{T}_{x^n y^n}$ in the discrete proof [9, Section IV-A] is replaced by $\hat{\mathbf{S}}(\mathbf{x}^n)$, but this replacement needs an operational justification. We first specify the common and private codebooks, the three

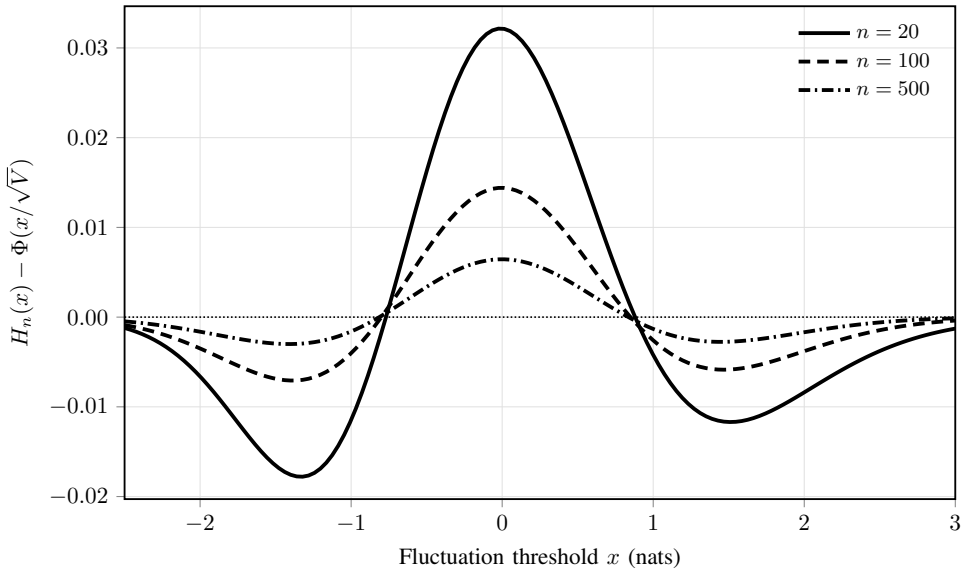


Fig. 3. Finite-blocklength CDF of the linearized source statistic minus its Gaussian limit, evaluated by (104). The tuple has $\rho = 0.6$, $D_1 = D_2 = 0.2$, unit marginal variances, and $\lambda_1^* = \lambda_2^* = 0.8$, so $V = 0.68$. Different dark line styles identify $n = 20, 100, 500$. These are exact source-statistic distributions evaluated numerically, not error probabilities of the covering code or finite-blocklength operational bounds.

encoding maps through their aggregate f, and both decoders. Lemmas 2 and 3 then justify the selected covariance designs and uniform shell covering. Lemma 4 combines the cell codes into one fixed-size code and gives its error event. At nonzero common rate, a scalar covariance expansion evaluates that event. At the corner with $L_0 > 0$, Lemma 5 and a bivariate limit replace that expansion. When $L_0 = 0$, the separate private-ball code below attains the same joint limit without a common message.

A. Codebook Generation, Encoding, and Decoding

We specify the one-shell and cell-indexed codes first, followed by a simpler private-ball code for $L_0 = 0$ at the corner. All codebooks, representatives, and tie rules are fixed and known to the encoder and both decoders before their performance is analyzed.

Recall the empirical matrix map (3). Fix $C_0 > 0$. For $\mathbf{S} \in \mathbb{S}_{++}^2$, put

$$\mathcal{T}_n(\mathbf{S}) := \left\{ \mathbf{x}^n \in \mathbb{R}^{2\times n} : \left\| \hat{\mathbf{S}}(\mathbf{x}^n) - \mathbf{S} \right\| \le C_0/n \right\}. \tag{105}$$

For $n \ge 3$, let $s_n := \sqrt{n}/\log n$. The design parameters are $(\mathbf{S}, \mathbf{K}) \in (\mathbb{S}_{++}^2)^2$ and $\kappa > 0$ satisfying

$$\kappa \mathbf{I}_2 \preceq \mathbf{K} \preceq \mathbf{S} \preceq \kappa^{-1} \mathbf{I}_2. \tag{106}$$

Choose the principal square root $\mathbf{B} := (\mathbf{S} - \mathbf{K})^{1/2}$. Generate independent normalized blocks $\mathbf{u}^n(m_0) \in \mathbb{R}^{2\times n}$ with independent $\mathcal{N}(\mathbf{0}, \mathbf{I}_2)$ columns. Their common auxiliary blocks are $\mathbf{w}^n(m_0) := \mathbf{B}\mathbf{u}^n(m_0)$. Thus $\mathbf{W} = \mathbf{B}\mathbf{U}$ is the Gaussian common auxiliary, whereas $\mathbf{U} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_2)$ is only its nonsingular innovation representation. Set

$$M_0 := \left\lceil \exp\left\{ \frac{n}{2} \log \frac{|\mathbf{S}|}{|\mathbf{K}|} + \frac{s_n}{8} \right\} \right\rceil, \tag{107}$$

$$\mathcal{U}_n := \{\mathbf{u}^n(m_0) : m_0 \in [1 : M_0]\}, \tag{108}$$

$$\mathcal{W}_n := \{\mathbf{w}^n(m_0) : m_0 \in [1 : M_0]\}, \tag{109}$$

$$Q_{0,n} := \left(\mathcal{N}(\mathbf{0}, \mathbf{I}_2)^{\otimes n}\right)^{\otimes M_0}. \tag{110}$$

Here $Q_{0,n}$ is the distribution of the normalized codebook $\mathcal{U}_n$, not the source distribution. Its image under $\mathbf{B}$ determines $\mathcal{W}_n$. The encoder and decoders use the same fixed realization; $\mathbf{w}^n(m_0)$ is its common reconstruction block.

For the private codebooks, set

$$\chi_\kappa := 2 + 2\sqrt{\kappa^{-1} + 1}, \tag{111}$$

$$\xi_{j,n} := \sqrt{nK_{jj} + \chi_\kappa s_n}, \qquad j \in [1:2]. \tag{112}$$

Let $C_{\mathrm{cov}} \geq 1$ depend only on κ, D_1, D_2, as supplied by Lemma 1. Choose deterministic, ordered offset sets $\mathcal{Y}_{j,n} := \{v_j^n(m_j) : m_j \in [1:M_j]\}$ such that

$$\mathcal{B}_n(\mathbf{0}, \xi_{j,n}) \subseteq \bigcup_{m_j \in [1:M_j]} \mathcal{B}_n(v_j^n(m_j), \sqrt{nD_j}), \tag{113}$$

$$M_j \leq C_{\mathrm{cov}} n^4 \max\left\{1, \left(\frac{\xi_{j,n}^2}{nD_j}\right)^{n/2}\right\}, \qquad j \in [1:2]. \tag{114}$$

Lemma 1 supplies these sets uniformly, including sequences of radius ratios tending to one; a single zero offset suffices when $\xi_{j,n}^2 \leq nD_j$. The same offsets are translated by each common reconstruction, so the private alphabet does not acquire a factor of M_0. This is the Euclidean covering idea used in [16, Lemma 16], but only a uniform polynomial prefactor, proved here, is required. The codebook choices precede all source and covering-probability analysis.

For a fixed realization of $\mathcal{U}_n$, define the eligible common indices and the associated private indices by

$$\mathcal{A}_{0,n}(\mathbf{x}^n) := \left\{m_0 \in [1:M_0] : \|(\mathbf{x}^n)_j - (\mathbf{B}\mathbf{u}^n(m_0))_j\|^2 \leq \xi_{j,n}^2,\ j \in [1:2]\right\}, \tag{115}$$

$$\mathcal{A}_{j,n}(\mathbf{x}^n, m_0) := \left\{m_j \in [1:M_j] : \left\|(\mathbf{x}^n)_j - (\mathbf{B}\mathbf{u}^n(m_0))_j - v_j^n(m_j)\right\|^2 \leq nD_j\right\}, \tag{116}$$

$$\mathcal{H}_n := \{\mathbf{x}^n \in \mathbb{R}^{2\times n} : \mathcal{A}_{0,n}(\mathbf{x}^n) \neq \emptyset\}. \tag{117}$$

For $\mathbf{x}^n \in \mathcal{H}_n$, let $\mu_0(\mathbf{x}^n)$ be the smallest index in $\mathcal{A}_{0,n}(\mathbf{x}^n)$ and let $\mu_j(\mathbf{x}^n)$ be the smallest index in $\mathcal{A}_{j,n}(\mathbf{x}^n, \mu_0(\mathbf{x}^n))$. The latter sets are nonempty by the private codebook design. Define

$$f^{\mathbf{S},\mathbf{K}}(\mathbf{x}^n) := \begin{cases} (\mu_0(\mathbf{x}^n), \mu_1(\mathbf{x}^n), \mu_2(\mathbf{x}^n)), & \mathbf{x}^n \in \mathcal{H}_n, \\ (1,1,1), & \mathbf{x}^n \in \mathcal{H}_n^{\mathrm{c}}, \end{cases} \tag{118}$$

$$\phi_j^{\mathbf{S},\mathbf{K}}(m_0, m_j) := (\mathbf{B}\mathbf{u}^n(m_0))_j + v_j^n(m_j), \qquad j \in [1:2]. \tag{119}$$

The fallback tuple makes the encoder defined on every block. The encoder tests only residual energies and distortion balls; it does not draw a posterior auxiliary or require an information-density test. Appendix C uses those posterior statistics only to prove that a common-codebook realization can be selected with $\mathcal{T}_n(\mathbf{S}) \subseteq \mathcal{H}_n$. The deterministic code uses that selected realization. Each decoder needs only its common and private indices.

Fix design targets $(r_0, r_1, r_2) \in \mathbb{R}_+^3$. Choose compact neighborhoods $\mathcal{K}_0 \subset \mathrm{int}(\mathcal{K}) \subset \mathbb{S}_{++}^2$ of $\mathbf{\Sigma}$ such that, for every $\mathbf{S} \in \mathcal{K}$,

$$S_{jj} > D_j \quad (j \in [1:2]), \qquad \rho(\mathbf{S}) \in [\rho/2, (1+\rho)/2]. \tag{120}$$

Round the three independent entries to nearest multiples of $1/n$, with a fixed tie rule, and denote the quantizer by q_n. For $\mathbf{S} \in \mathcal{K}_0$ and sufficiently large n, the representative lies in $\mathcal{K}$ and

$$\|q_n(\mathbf{S}) - \mathbf{S}\| \leq 1/n. \tag{121}$$

The Frobenius norm counts the off-diagonal error twice. Take $C_0 \geq 1$ so that each cell is contained in its shell. To define the analysis statistics on every source block, use the safe extension

$$\bar{q}_n(\mathbf{S}) := \begin{cases} q_n(\mathbf{S}), & \mathbf{S} \in \mathcal{K}_0, \\ \mathbf{\Sigma}, & \mathbf{S} \in \mathbb{S}_+^2 \setminus \mathcal{K}_0, \end{cases} \tag{122}$$

$$\mathbf{S}_n := \bar{q}_n(\hat{\mathbf{\Sigma}}_n) \in \mathcal{K}. \tag{123}$$

These definitions apply once $q_n(\mathcal{K}_0) \subset \mathcal{K}$; for the finitely many smaller blocklengths take $\bar{q}_n \equiv \mathbf{\Sigma}$ and use the fallback code. The raw entrywise quantizer is used only inside $\mathcal{K}_0$. The safe extension changes no retained-cell code or typical-set event, but ensures that all subsequent covariance determinants, logarithms, and boundary values are defined globally. Enumerate the grid representatives as $\mathbf{S}_q$, $q \in [1:N_n]$, and set

$$\mathcal{T}_{n,q} := \{\mathbf{x}^n \in \mathbb{R}^{2\times n} : \hat{\mathbf{S}}(\mathbf{x}^n) \in \mathcal{K}_0,\ q_n(\hat{\mathbf{S}}(\mathbf{x}^n)) = \mathbf{S}_q\}, \tag{124}$$

$$\mathcal{J}_n := \{q \in [1:N_n] : \mathsf{R}_0(\mathbf{S}_q; r_1, r_2) \leq r_0\}, \tag{125}$$

$$\mathcal{T}_n^\circ := \bigcup_{q \in \mathcal{J}_n} \mathcal{T}_{n,q}. \tag{126}$$

For each retained cell $q \in \mathcal{J}_n$, let $\mathbf{K}_q := \mathbf{K}^*(\mathbf{S}_q; r_1, r_2)$ be the unique optimizer of (27), and use the one-shell construction above with $(\mathbf{S}_q, \mathbf{K}_q)$. Existence and uniform positive definiteness follow from Lemma 2; uniqueness follows from strict concavity

of $\log|\mathbf{K}|$ on the convex feasible set. Appendix A-B evaluates the optimizer in (249)–(250) and verifies the formulas. Write f_q, $\phi_{j,q}$, and $M_{j,q}$ for the cell encoder, decoders, and sizes. The covering lemma below supplies a deterministic code for each of these finitely many covariance pairs.

The global common alphabet reserves one fallback symbol and otherwise specifies the cell and its local common index. Define

$$\mathcal{M}_{0,n} := \{0\} \cup \bigcup_{q\in\mathcal{J}_n} (\{q\} \times [1 : M_{0,q}]), \tag{127}$$

$$M_0 := |\mathcal{M}_{0,n}|, \tag{128}$$

$$M_j := \max\big(\{1\} \cup \{M_{j,q} : q \in \mathcal{J}_n\}\big), \qquad j \in [1 : 2]. \tag{129}$$

Fix a bijection $\iota_n : \mathcal{M}_{0,n} \to [1 : M_0]$. If $\mathbf{x}^n \in \mathcal{T}_{n,q}$ for $q \in \mathcal{J}_n$, compute $(m_0, m_1, m_2) := f_q(\mathbf{x}^n)$ and transmit

$$f(\mathbf{x}^n) := (\iota_n(q, m_0), m_1, m_2). \tag{130}$$

For $\mathbf{x}^n \notin \mathcal{T}_n^\circ$, transmit $(\iota_n(0), 1, 1)$. Thus the cell index is carried only by the common message. For $q \in \mathcal{J}_n$, $m_0 \in [1 : M_{0,q}]$, and $m_j \in [1 : M_{j,q}]$, Decoder j outputs

$$\phi_j(\iota_n(q, m_0), m_j) := \phi_{j,q}(m_0, m_j). \tag{131}$$

It outputs the zero vector on the fallback common symbol and on unused padded private indices. When $\mathcal{J}_n = \emptyset$, this is the one-message fallback code. These definitions specify every encoding and decoding operation. We now analyze their existence, message sizes, measurability, and error probability.

At $\mathbf{R}_{\mathrm{c}}^*$ with $L_0 = 0$, the following simpler code will replace the cell-indexed construction. Fix $(L_1, L_2) \in \mathbb{R}^2$ and set $M_{0,n} = 1$. For $n \geq 3$, choose ordered private reproduction sets $\mathcal{Y}_{j,n}^{(0)} = \{v_{j,n}^{(0)}(m) : m \in [1 : M_{j,n}^{(0)}]\}$ covering the fixed balls of radii

$$\xi_{j,n}^{(0)} := \sqrt{n\Sigma_{jj}}\, e^{L_j/\sqrt{n}}, \qquad j \in [1 : 2]. \tag{132}$$

Specifically, apply the Euclidean covering bound in Lemma 1 with covering radius $\sqrt{nD_j}$; the ratio is bounded uniformly in $n \geq 3$ for these fixed parameters. For every input block define

$$\mathcal{A}_{j,n}^{(0)}(\mathbf{x}^n) := \{m \in [1 : M_{j,n}^{(0)}] : \left\|(\mathbf{x}^n)_j - v_{j,n}^{(0)}(m)\right\|^2 \leq nD_j\}, \tag{133}$$

$$f_0^{(0)}(\mathbf{x}^n) := 1, \tag{134}$$

$$f_j^{(0)}(\mathbf{x}^n) := \begin{cases} \min \mathcal{A}_{j,n}^{(0)}(\mathbf{x}^n), & \mathcal{A}_{j,n}^{(0)}(\mathbf{x}^n) \neq \emptyset, \\ 1, & \mathcal{A}_{j,n}^{(0)}(\mathbf{x}^n) = \emptyset, \end{cases} \tag{135}$$

$$\phi_j^{(0)}(1, m) := v_{j,n}^{(0)}(m), \qquad j \in [1 : 2]. \tag{136}$$

For $n = 1, 2$, use singleton alphabets and zero reconstructions. Each private encoder depends only on its own row; the joint encoder may implement both. No empirical-covariance index is sent. The fixed finite covers and smallest-index rule make the maps Borel, and the fallback defines them on every block. Section IV-C proves their rate and joint-success bounds.

B. Covering Bounds and Rate Budgets

For the code of Section IV-A, the following lemmas control the private coverings, selected residual covariances, shell coverings, and combined message sizes and error event. All use the fixed distortions and compact covariance neighborhoods specified there.

Lemma 1 (Uniform Euclidean ball covering)**.** *Fix $T \geq 1$. There is $C_T \geq 1$ such that, for every integer $n \geq 3$, $r > 0$, and $R \in [0, Tr]$, a finite set $\mathcal{V} \subset \mathbb{R}^n$ satisfies*

$$\mathcal{B}_n(\mathbf{0}, R) \subseteq \bigcup_{\mathbf{v}\in\mathcal{V}} \mathcal{B}_n(\mathbf{v}, r), \tag{137}$$

$$|\mathcal{V}| \leq C_T n^4 \max\{1, (R/r)^n\}. \tag{138}$$

When $R \leq r$, one may take $\mathcal{V} = \{\mathbf{0}\}$. The constant and the threshold $n \geq 3$ do not depend on R/r.

Appendix C-A proves this weaker polynomial version directly by a radial grid, spherical caps, and finite nets. It avoids inferring uniformity at $R/r \downarrow 1$ from a fixed-ratio asymptotic assertion. In the construction above,

$$\frac{\xi_{j,n}}{\sqrt{nD_j}} \leq \sqrt{\frac{\kappa^{-1} + \chi_\kappa}{\min\{D_1, D_2\}}}, \qquad n \geq 3, \quad j \in [1 : 2]. \tag{139}$$

Thus one fixed T and one C_{cov} apply to all shells and blocklengths, including zero private first-order rates.

Lemma 2 (Compactness of residual optimizers)**.** *Fix $(D_1, D_2) \in \mathbb{R}_+^2$ with $\min\{D_1, D_2\} > 0$ and a compact $\mathcal{K} \subset \mathbb{S}_{++}^2$. For every $\mathbf{S} \in \mathcal{K}$ and $(r_1, r_2) \in \mathbb{R}_+^2$, the minimum in* (27) *is attained. There is $\kappa > 0$, independent of $\mathbf{S}$ and the private rates, such that every optimizer satisfies*

$$\kappa \mathbf{I}_2 \preceq \mathbf{K}^* \preceq \mathbf{S} \preceq \kappa^{-1} \mathbf{I}_2. \tag{140}$$

Appendix C-A proves this uniform bound by comparing the optimizer with a small diagonal covariance.

Lemma 3 (Covariance-shell covering)**.** *Fix $(D_1, D_2, C_0) \in \mathbb{R}_+^3$ with $\min\{D_1, D_2, C_0\} > 0$ and $\kappa > 0$. Uniformly over all pairs satisfying*

$$\kappa \mathbf{I}_2 \preceq \mathbf{K} \preceq \mathbf{S} \preceq \kappa^{-1} \mathbf{I}_2, \tag{141}$$

there are constants $C < \infty$ and $n_0 \in \mathbb{N}$ such that, for every $n \geq n_0$, a deterministic Gray–Wyner code covers every sequence in $\mathcal{T}_n(\mathbf{S})$ within both distortions and satisfies

$$\log M_0 \leq \frac{n}{2} \log \frac{|\mathbf{S}|}{|\mathbf{K}|} + C s_n, \tag{142}$$

$$\log M_j \leq \frac{n}{2} \left[\log \frac{K_{jj}}{D_j} \right]^+ + C s_n + C \log n, \qquad j \in [1:2]. \tag{143}$$

No lower bound is imposed on the nonzero eigenvalues of $\mathbf{S} - \mathbf{K}$.

The constants in Lemma 3 are uniform over the covariance family, while the deterministic code is selected separately for each fixed pair $(\mathbf{S}, \mathbf{K})$. The operational construction uses only finitely many such selections at each blocklength. Appendix C proves the lemma using a conditional small-ball probability bounded below by $c(\log n)^{-3}$ for a fixed $c > 0$, uniformly over that family. The common-rate margin makes the failure probability small enough to cover a finite net of the entire shell. Extending from that net and adding the private coverings costs $O(s_n + \log n) = o(\sqrt{n})$ nats. The argument allows arbitrarily small nonzero eigenvalues of the common covariance.

Lemma 4 (Operational shell bound)**.** *There is $C < \infty$ such that, with $h_n := C(s_n + \log n)$, every choice of target rates $(r_0, r_1, r_2) \in \mathbb{R}_+^3$ admits, for all sufficiently large n, a code with*

$$\log M_j \leq n r_j + h_n, \qquad j \in [0:2], \tag{144}$$

and

$$\varepsilon_n \leq P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \notin \mathcal{K}_0\} + P_{XY}^n\big\{\hat{\mathbf{\Sigma}}_n \in \mathcal{K}_0,\ \mathsf{R}_0(\mathbf{S}_n; r_1, r_2) > r_0\big\}. \tag{145}$$

The constant is uniform over the rates in this statement.

Appendix C-C counts the combined message alphabets and proves this error bound for the code in Section IV-A.

The zero common rate analysis additionally requires the following local expansion of Υ_ρ. Recall ψ_ρ from (61) and the Gaussian vector $\mathbf{Z}$ with distribution $P_{\mathbf{Z}}$ from (62).

Lemma 5 (Directional expansion and positive levels)**.** *Fix $\rho \in (0,1)$. For every $(M, H) \in \mathbb{R}_+^2$, there are constants $C_{M,H} < \infty$ and $t_{M,H} > 0$ such that*

$$\sup_{\substack{0 \leq a,b \leq M \\ |h| \leq H}} |\Upsilon_{\rho+th}(ta, tb) - t\psi_\rho(a,b)| \leq C_{M,H} t^2, \qquad 0 < t < t_{M,H}. \tag{146}$$

For any $(L_1, L_2) \in \mathbb{R}^2$, define $T : \mathbb{R}^2 \to \mathbb{R}_+$ by

$$T(\mathbf{z}) := \psi_\rho([z_1 - L_1]^+, [z_2 - L_2]^+), \qquad \mathbf{z} = (z_1, z_2)^{\mathsf{T}}. \tag{147}$$

The statistic $T(\mathbf{Z})$ has no atom at a positive level. Its mass at zero is

$$P_{\mathbf{Z}}\{T(\mathbf{Z}) = 0\} = P_{\mathbf{Z}}\{Z_1 \leq L_1,\ Z_2 \leq L_2\}. \tag{148}$$

Appendix B proves the uniform expansion, the Lipschitz properties of ψ_ρ, and the assertion about positive levels.

Fix an offset satisfying the one-sided constraints in Theorem 1. For the cell-indexed construction, let $b_n = (\log n)^{-1/4}$ and choose

$$r_{0,n} := R_0^* + (L_0 + b_n)/\sqrt{n}, \tag{149}$$

$$r_{j,n} := \max\{0, R_j^* + L_j/\sqrt{n}\}, \qquad j \in [1:2]. \tag{150}$$

These targets are eventually nonnegative; the maximum is inactive at positive private rates and equals $L_j/\sqrt{n}$ when $R_j^* = 0$. Lemma 4 gives

$$\limsup_{n\to\infty} \frac{\log M_{j,n} - nR_j^*}{\sqrt{n}} \le L_j + \limsup_{n\to\infty} \left(b_n + \frac{h_n}{\sqrt{n}}\right) = L_j, \qquad j \in [0:2], \tag{151}$$

$$\varepsilon_n \le P_{XY}^n \left\{\hat{\mathbf{\Sigma}}_n \in \mathcal{K}_0, \sqrt{n}[\mathsf{R}_0(\mathbf{S}_n; r_{1,n}, r_{2,n}) - R_0^*] > L_0 + b_n\right\} + o(1). \tag{152}$$

where Eq. (151) uses the alphabet bound and $h_n = o(\sqrt{n})$, and Eq. (152) substitutes the targets into (145) and uses $P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \notin \mathcal{K}_0\} \to 0$. We now evaluate this operational event. For $L_0 = 0$ at the corner, the previously defined private-only code will instead suffice.

C. Evaluation of the Coding Error

For $R_0^* > 0$, Proposition 2 gives

$$\mathsf{R}_0(\mathbf{S}; r_1, r_2) = R_0^* + \langle \mathbf{G}, \mathbf{S} - \mathbf{\Sigma}\rangle - \sum_{j\in[1:2]} \lambda_j^*(r_j - R_j^*) + o\left(\|\mathbf{S} - \mathbf{\Sigma}\| + \sum_{j\in[1:2]} |r_j - R_j^*|\right). \tag{153}$$

By (75), the statistic A_n of (102) equals the normalized sum of independent centered variables $\jmath_{XY}(X_i, Y_i) - R_0^*$. They have variance $V > 0$ and finite third moment, so $\mathsf{A}_n \Rightarrow \mathcal{N}(0, V)$ under P_{XY}^n.

Define the global remainder by

$$E_n := \sqrt{n}[\mathsf{R}_0(\mathbf{S}_n; r_{1,n}, r_{2,n}) - R_0^*] - \mathsf{A}_n + \lambda_1^* L_1 + \lambda_2^* L_2. \tag{154}$$

It is finite because the safe representative lies in $\mathcal{K}$. The scaled empirical covariance is tight by its bounded second moments. On any bounded scaled-covariance event, (153) has a uniformly vanishing scaled remainder, and quantization contributes $O(n^{-1/2})$. These events eventually lie in $\mathcal{K}_0$. Thus

$$P_{XY}^n\{|E_n| > \zeta\} \longrightarrow 0, \qquad \zeta > 0. \tag{155}$$

No magnitude estimate for E_n outside $\mathcal{K}_0$ is required. Set $\ell = L_0 + \lambda_1^* L_1 + \lambda_2^* L_2$. For fixed $\zeta > 0$,

$$\begin{aligned} \varepsilon_n &\le P_{XY}^n\{\mathsf{A}_n + E_n > \ell + b_n\} + o(1) \\ &\le P_{XY}^n\{\mathsf{A}_n > \ell - \zeta\} + P_{XY}^n\{|E_n| > \zeta\} + o(1), \end{aligned} \tag{156}$$

$$\limsup_{n\to\infty} \varepsilon_n \le \lim_{\zeta\downarrow 0} \mathrm{Q}\left(\frac{\ell - \zeta}{\sqrt{V}}\right) = \mathrm{Q}\left(\frac{L_0 + \lambda_1^* L_1 + \lambda_2^* L_2}{\sqrt{V}}\right). \tag{157}$$

where Eq. (156) uses (152), the remainder identity, $b_n \ge 0$, and the union bound. Equation (157) takes $n \to \infty$ at fixed ζ using the central limit theorem and (155), then sends $\zeta \downarrow 0$. This proves case (i); only continuous differentiability, not bounded second derivatives at interfaces, was used.

We next treat zero common rate. Let $\mathbf{R}^* = \mathbf{R}_{\mathrm{c}}^*$. Define the powers, correlation, and marginal fluctuations by

$$\hat{\sigma}_{j,n}^2 := [\hat{\mathbf{\Sigma}}_n]_{jj}, \qquad j \in [1:2], \tag{158}$$

$$\hat{\rho}_n := \frac{[\hat{\mathbf{\Sigma}}_n]_{12}}{\hat{\sigma}_{1,n}\hat{\sigma}_{2,n}}, \tag{159}$$

$$Z_{j,n} := \frac{\sqrt{n}}{2} \log \frac{\hat{\sigma}_{j,n}^2}{\Sigma_{jj}}, \qquad j \in [1:2]. \tag{160}$$

On the null set of vanishing power, assign these statistics and the deficits below the value zero. The logarithmic Taylor expansion and the multivariate central limit theorem give, under P_{XY}^n,

$$Z_{j,n} = \frac{1}{2\Sigma_{jj}\sqrt{n}} \sum_{i\in[1:n]} ([\mathbf{X}_i]_j^2 - \Sigma_{jj}) + E'_{j,n}, \tag{161}$$

$$P_{XY}^n\{|E'_{j,n}| > \zeta\} \longrightarrow 0, \qquad \zeta > 0, \tag{162}$$

$$\mathbf{Z}_n := (Z_{1,n}, Z_{2,n})^{\mathsf{T}} \Rightarrow \mathbf{Z}. \tag{163}$$

where Eq. (162) uses tightness of the scaled powers and the quadratic remainder. In (163), the covariance is (62) because $\mathsf{Var}([\mathbf{X}]_j^2) = 2\Sigma_{jj}^2$ and $\mathsf{Cov}(X^2, Y^2) = 2\rho^2\sigma_X^2\sigma_Y^2$. Differentiability of correlation also makes $\sqrt{n}(\hat{\rho}_n - \rho)$ tight.

For the design rates (150), set $\Delta_{j,n} = [\frac{1}{2}\log(\hat{\sigma}^2_{j,n}/D_j) - r_{j,n}]^+$. Both private first-order rates are positive, and therefore, for sufficiently large n,

$$\sqrt{n}\,\Delta_{j,n} = \left[\frac{\sqrt{n}}{2}\log\frac{\hat{\sigma}^2_{j,n}}{\Sigma_{jj}} - L_j\right]^+ = [Z_{j,n} - L_j]^+. \tag{164}$$

where Eq. (164) substitutes $R^*_j = \frac{1}{2}\log(\Sigma_{jj}/D_j)$. For $B > 0$, define

$$\mathcal{O}_n(B) := \left\{\max_{j\in[1:2]} |Z_{j,n}| \le B, \sqrt{n}|\hat{\rho}_n - \rho| \le B\right\}. \tag{165}$$

Its complement has arbitrarily small limiting probability as $B \to \infty$. At fixed B, it eventually lies in $\{\hat{\mathbf{\Sigma}}_n \in \mathcal{K}_0\}$, since the power ratios are $e^{2Z_{j,n}/\sqrt{n}}$. Hence safe and raw representatives agree there. Uniformly on this event,

$$\begin{aligned}\sqrt{n}\,\mathsf{R}_0(\mathbf{S}_n; r_{1,n}, r_{2,n}) &= \sqrt{n}\,\Upsilon_{\rho(\mathbf{S}_n)}(\alpha(\mathbf{S}_n; r_{1,n}), \beta(\mathbf{S}_n; r_{2,n})) \\ &= \psi_\rho(\sqrt{n}\,\alpha(\mathbf{S}_n; r_{1,n}), \sqrt{n}\,\beta(\mathbf{S}_n; r_{2,n})) + O_B(n^{-1/2}) \\ &= \psi_\rho([Z_{1,n} - L_1]^+, [Z_{2,n} - L_2]^+) + O_B(n^{-1/2}).\end{aligned} \tag{166}$$

where Eq. (166) uses Yu's representation, Lemma 5 at $t = n^{-1/2}$, and (164). Quantization changes the scaled deficits by $O(n^{-1/2})$, controlled by (260); correlation fluctuations contribute only to the remainder.

For $L_0 > 0$, recall T from (147). The coding event (152) gives

$$\varepsilon_n \le P^n_{XY}\{T(\mathbf{Z}_n) > L_0 + b_n - C_B n^{-1/2}\} + P^n_{XY}(\mathcal{O}_n(B)^{\mathrm{c}}) + o(1), \tag{167}$$

$$\begin{aligned}\limsup_{n\to\infty} \varepsilon_n &\le \lim_{B\to\infty}\left[P_{\mathbf{Z}}\{T(\mathbf{Z}) > L_0\} + \limsup_{n\to\infty} P^n_{XY}(\mathcal{O}_n(B)^{\mathrm{c}})\right] \\ &= P_{\mathbf{Z}}\{\psi_\rho([Z_1 - L_1]^+, [Z_2 - L_2]^+) > L_0\}.\end{aligned} \tag{168}$$

where Eq. (168) first takes $n \to \infty$ at fixed B, using (163) and the absence of positive atoms in Lemma 5, then removes localization by tightness.

For $L_0 = 0$, use the private-only code of Section IV-A. Its fixed offsets make $\xi^{(0)}_{j,n}/\sqrt{nD_j}$ bounded and greater than one for large n, so Lemma 1 gives

$$\begin{aligned}\log M^{(0)}_{j,n} &\le \frac{n}{2}\log\frac{(\xi^{(0)}_{j,n})^2}{nD_j} + 4\log n + C \\ &= nR^*_j + \sqrt{n}L_j + 4\log n + C.\end{aligned} \tag{169}$$

where Eq. (169) substitutes (132). Thus the common offset is zero and each private offset is at most L_j. The covered-ball event is $\mathcal{A}^{(0)}_n := \{\mathbf{x}^n \in \mathbb{R}^{2\times n} : \|(\mathbf{x}^n)_j\|^2 \le n\Sigma_{jj}e^{2L_j/\sqrt{n}},\ j \in [1:2]\}$, and

$$1 - \varepsilon^{(0)}_n \ge P^n_{XY}(\mathcal{A}^{(0)}_n) = P^n_{XY}\{Z_{1,n} \le L_1, Z_{2,n} \le L_2\}, \tag{170}$$

$$\liminf_{n\to\infty}(1 - \varepsilon^{(0)}_n) \ge P_{\mathbf{Z}}\{Z_1 \le L_1, Z_2 \le L_2\} = P_{\mathbf{Z}}\{T(\mathbf{Z}) = 0\}. \tag{171}$$

where Eq. (170) uses the private covers and logarithmic power identity. Equation (171) applies (163) at a rectangle with zero boundary probability and (148). This completes case (ii) and the direct inclusion in (81) without any common header.

V. Converse Proof

We prove the converse part of Theorem 1. At nonzero common rate, we identify the entropy support and dispersion in the published Gaussian converse, with a separate scalar argument for a zero supporting weight. At zero common rate, a joint image-size bound is needed before the bivariate source limit can be taken. Fix an arbitrary code sequence satisfying (51). If $R^*_j = 0$, the inequality $M_{j,n} \ge 1$ forces $L_j \ge 0$. For every fixed $\gamma > 0$, the remaining rate constraints give the eventual envelope

$$\log M_{j,n} \le nR^*_j + \sqrt{n}(L_j + \gamma), \qquad j \in [0:2]. \tag{172}$$

We use this one-sided statement throughout and send γ to zero only after the blocklength limit.

A. Nonzero Common Rate

Fix a nonzero common rate Pareto-minimal boundary tuple whose slopes satisfy $\boldsymbol{\lambda}^* = (\lambda_1^*, \lambda_2^*) \in (0,1]^2$. The existing Gaussian converse uses the entropy support

$$\underline{F}_{\boldsymbol{\lambda}^*}(\mathbf{S}) := \inf_{\substack{\mathbf{K}\in\mathbb{S}^2_{++} \\ \mathbf{K}\preceq\mathbf{S}}} \left\{ \frac{1}{2}\log\frac{|\mathbf{S}|}{|\mathbf{K}|} + \sum_{j\in[1:2]} \frac{\lambda_j^*}{2}\log\frac{K_{jj}}{D_j} \right\}, \qquad \mathbf{S}\in\mathbb{S}^2_{++}. \tag{173}$$

The logarithms are not clipped at zero; this entropy support differs from the operational support. Keep the true covariance $\boldsymbol{\Sigma}$ fixed and define

$$\varphi_{\boldsymbol{\lambda}^*,\boldsymbol{\Sigma}}(\mathbf{S}) := -\underline{F}_{\boldsymbol{\lambda}^*}(\mathbf{S}) + \frac{1}{2}\log|\mathbf{S}| - \frac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}). \tag{174}$$

A matrix $\mathbf{B}_{\boldsymbol{\lambda}^*} \in \mathbb{S}^2$ is a *supergradient* of this function at $\boldsymbol{\Sigma}$ if

$$\varphi_{\boldsymbol{\lambda}^*,\boldsymbol{\Sigma}}(\mathbf{S}) \le \varphi_{\boldsymbol{\lambda}^*,\boldsymbol{\Sigma}}(\boldsymbol{\Sigma}) + \langle \mathbf{B}_{\boldsymbol{\lambda}^*}, \mathbf{S}-\boldsymbol{\Sigma}\rangle, \qquad \mathbf{S}\in\mathbb{S}^2_{++}. \tag{175}$$

Thus its affine plane bounds the function from above. For such a matrix, define the centered quadratic function

$$g_{\boldsymbol{\lambda}^*}(\mathbf{x}) := \mathbf{x}^{\mathsf{T}}\mathbf{B}_{\boldsymbol{\lambda}^*}\mathbf{x} - \operatorname{tr}(\mathbf{B}_{\boldsymbol{\lambda}^*}\boldsymbol{\Sigma}), \qquad \mathbf{x}\in\mathbb{R}^2. \tag{176}$$

Liu et al. [11, Proposition 5, Eqs. (64)–(74)] supplied concavity and the correspondence between matrix and quadratic functional supergradients. Equation (175) fixes the sign convention; centering removes the additive constant. We identify the support and fluctuation at the selected tuple as follows.

Lemma 6 (Entropy support at the selected boundary)**.** *Let $\mathbf{R}^*$ be a Pareto-minimal boundary tuple with $R_0^* > 0$ and $(\lambda_1^*, \lambda_2^*) \in \mathbb{R}_+^2$ satisfying $\min\{\lambda_1^*, \lambda_2^*\} > 0$, where the weights are defined by* (54)–(55)*. Then*

$$\underline{F}_{\boldsymbol{\lambda}^*}(\boldsymbol{\Sigma}) = R_0^* + \lambda_1^* R_1^* + \lambda_2^* R_2^*, \tag{177}$$

$$\nabla_{\mathbf{S}}\underline{F}_{\boldsymbol{\lambda}^*}(\mathbf{S})|_{\mathbf{S}=\boldsymbol{\Sigma}} = \mathbf{G}. \tag{178}$$

The unique supergradient in (175) *is $\mathbf{B}_{\boldsymbol{\lambda}^*} = -\mathbf{G}$. Consequently,*

$$g_{\boldsymbol{\lambda}^*}(\mathbf{x}) = -[\jmath_{XY}(x,y) - R_0^*], \qquad \mathbf{x} = (x,y)^{\mathsf{T}}, \tag{179}$$

$$\mathsf{Var}_{P_{XY}}[g_{\boldsymbol{\lambda}^*}(\mathbf{X})] = V. \tag{180}$$

Appendix A-B proves this lemma, including the case of a zero private first-order rate. The proof checks equality at the selected covariance and a touching inequality nearby; it does not identify the clipped operational support with the entropy support on a whole neighborhood.

Lemma 7 (Positive-weight Gaussian converse, specialized)**.** *Let $\mathbf{R}^*$ be a nonzero-common-rate Pareto-minimal boundary tuple with $\min\{\lambda_1^*, \lambda_2^*\} > 0$, and let a sequence of codes satisfy the second-order rate envelopes in* (172) *eventually for every $\gamma > 0$. Then*

$$\liminf_{n\to\infty} \varepsilon_n \ge \mathrm{Q}\left(\frac{L_0 + \lambda_1^* L_1 + \lambda_2^* L_2}{\sqrt{V}}\right). \tag{181}$$

Proof. The only imported ingredient is [11, Theorem 17, with Proposition 5 and Eq. (190)]. In its notation the weighted displacement equals

$$A := \limsup_{n\to\infty} \frac{\log M_{0,n} + \lambda_1^* \log M_{1,n} + \lambda_2^* \log M_{2,n} - n\underline{F}_{\boldsymbol{\lambda}^*}(\boldsymbol{\Sigma})}{\sqrt{n}}. \tag{182}$$

Lemma 6 verifies the three identifications needed to apply that theorem: the support value is $R_0^* + \lambda_1^* R_1^* + \lambda_2^* R_2^*$, the centered functional subgradient is $-[\jmath_{XY}(X,Y) - R_0^*]$, and its variance is V. The source covariance is positive definite, the distortions and weights are positive, and the support is finite. The rate envelopes make A bounded above by a finite number. Suppose first that $A \in \mathbb{R}$. The cited theorem gives

$$\liminf_{n\to\infty} \varepsilon_n \ge \mathrm{Q}(A/\sqrt{V}). \tag{183}$$

For every $\gamma > 0$, the rate envelopes and (177) yield

$$A \le L_0 + \lambda_1^* L_1 + \lambda_2^* L_2 + (1 + \lambda_1^* + \lambda_2^*)\gamma. \tag{184}$$

Combining the last two displays, using that Q is decreasing, and then sending $\gamma \downarrow 0$ proves (181).

It remains to consider $A = -\infty$. Fix any $a \in \mathbb{R}$. By (182), the normalized weighted displacement is eventually at most $a-1$. The finite-threshold argument in the proof of [11, Theorem 17, Eqs. (193)–(197)], using the Gaussian distortion-ball volume estimate, therefore gives

$$\varepsilon_n \geq P_{XY}^n \left\{ \frac{1}{\sqrt{n}} \sum_{i\in[1:n]} g_{\boldsymbol{\lambda}^*}(\mathbf{X}_i) \leq -a \right\} - \exp\{-\sqrt{n} + O(\log n)\}, \tag{185}$$

$$\liminf_{n\to\infty} \varepsilon_n \geq \mathrm{Q}(a/\sqrt{V}). \tag{186}$$

where the logarithmic remainder in (185) depends only on the fixed distortions and weights. Equation (186) uses Lemma 6 and the central limit theorem for the independent centered summands with variance $V > 0$. Sending $a \downarrow -\infty$ gives $\liminf_{n\to\infty} \varepsilon_n \geq 1$, proving (181) in this remaining case. □

Lemma 7 matches (157) for every positive-weight supporting direction. The one-source faces require the separate argument below rather than a limiting sequence of positive weights.

On $\mathcal{D}_3$ and its interface, $R_0^* + R_1^* = R_X(P_X, D_1)$ and the normal is $(1, 1, 0)$. Decoder 1 has at most $M_{0,n}M_{1,n}$ reproductions. Its successful set lies in their distortion-ball union regardless of the encoder's observation of Y^n. For each x^n, choose the smallest aggregate index minimizing distortion over all $M_{0,n}M_{1,n}$ decoder reproductions. A finite minimum of continuous distortion functions with this tie rule defines a Borel encoder observing only X^n. If the original decoder succeeds for any y^n, this encoder also succeeds, so its excess-distortion probability is no larger. Thus the extra encoder observation Y^n cannot improve this scalar converse. The scalar Gaussian second-order converse [4]–[6] therefore applies, since

$$\begin{aligned} \frac{\log(M_{0,n}M_{1,n}) - nR_X(P_X, D_1)}{\sqrt{n}} &= \frac{\log M_{0,n} - nR_0^* + \log M_{1,n} - nR_1^*}{\sqrt{n}} \\ &\leq L_0 + L_1 + 2\gamma. \end{aligned} \tag{187}$$

where Eq. (187) uses the face identity and the eventual rate envelopes. The scalar converse gives $L_0 + L_1 + 2\gamma \geq \mathrm{Q}^{-1}(\varepsilon)/\sqrt{2}$ for every $\gamma > 0$. Hence

$$L_0 + L_1 \geq \frac{1}{\sqrt{2}} \mathrm{Q}^{-1}(\varepsilon). \tag{188}$$

This is (63) with $V = 1/2$; the other face follows by interchanging receivers. No limiting argument with positive Brascamp–Lieb weights is used.

B. Zero Common Rate

The main converse input is a finite-blocklength conditional success bound for every code, rather than separate necessary inequalities in fixed supporting directions. For $\mathbf{S} = [s_{jk}] \in \mathcal{K}$, let $Q_{\mathbf{S}} := \mathcal{N}(\mathbf{0}, \mathbf{S})$ and define

$$\mathcal{C}_n(\mathbf{S}) := \left\{ \mathbf{x}^n \in \mathbb{R}^{2\times n} : [\hat{\mathbf{S}}(\mathbf{x}^n)]_{jk} - s_{jk} \in [-1/(2n), 1/(2n)),\ 1 \leq j \leq k \leq 2 \right\}. \tag{189}$$

When $\mathbf{S}$ ranges over the nearest-grid representatives, these are disjoint cells. The conditional estimate below is valid for every $\mathbf{S} \in \mathcal{K}$, not only grid points. For an arbitrary code, define its two marginal rate deficits by

$$\delta_{j,n}(\mathbf{S}) := \left[\tfrac{1}{2}\log(s_{jj}/D_j) - n^{-1}\log M_{j,n}\right]^+, \qquad j \in [1:2]. \tag{190}$$

The successful set $\mathcal{A}_n$ is the set in (14) for that code.

Lemma 8 (Joint covariance-cell converse)**.** *Fix $\mathbf{\Sigma} \in \mathbb{S}_{++}^2$, $D_1, D_2 > 0$, and a compact $\mathcal{K} \subset \{\mathbf{S} \in \mathbb{S}_{++}^2 : S_{jj} > D_j\ (j \in [1:2]),\ S_{12} > 0\}$. There are $C \geq 1$, $k \geq 0$, and $n_0 \in \mathbb{N}$, depending only on these fixed parameters, such that every $(n, M_{0,n}, M_{1,n}, M_{2,n})$ code, every $n \geq n_0$, and every $\mathbf{S} \in \mathcal{K}$ satisfy*

$$P_{XY}^n(\mathcal{A}_n \mid \mathcal{C}_n(\mathbf{S})) \leq \min\left\{1, Cn^k \exp\left[\log M_{0,n} - n\psi_{\rho(\mathbf{S})}\big(\delta_{1,n}(\mathbf{S}), \delta_{2,n}(\mathbf{S})\big)\right]\right\}. \tag{191}$$

The constants do not depend on the code, its rates, its reconstruction centers, or the selected covariance.

Appendix D proves the lemma by combining the marginal image estimate with Gaussian hypercontractivity, summing over common messages, and using the polynomial covariance-cell mass and bounded likelihood-ratio oscillation. The proof optimizes over all feasible hypercontractive exponents before taking an asymptotic limit. The lemma is not an exact deterministic-shell rate characterization; it is the uniform joint converse needed for the corner fluctuation.

Set $A_{j,n}(\mathbf{S}) = \sqrt{n}\,\delta_{j,n}(\mathbf{S})$. For fixed $\gamma, \zeta > 0$, define

$$\mathcal{B}_{n,\gamma,\zeta} := \{\mathbf{S} \in \mathcal{K} : \psi_{\rho(\mathbf{S})}(A_{1,n}(\mathbf{S}), A_{2,n}(\mathbf{S})) > L_0 + \gamma + \zeta\}. \tag{192}$$

For $\mathbf{S}$ in this set, the common-rate envelope and homogeneity give

$$\begin{aligned} \log M_{0,n} - n\psi_{\rho(\mathbf{S})}(\delta_{1,n}(\mathbf{S}), \delta_{2,n}(\mathbf{S})) &\le \sqrt{n}[L_0 + \gamma - \psi_{\rho(\mathbf{S})}(A_{1,n}(\mathbf{S}), A_{2,n}(\mathbf{S}))] \\ &< -\zeta\sqrt{n}. \end{aligned} \tag{193}$$

Let $\mathcal{B}_n := \{\mathbf{x}^n \in \mathbb{R}^{2\times n} : \hat{\mathbf{S}}(\mathbf{x}^n) \in \mathcal{K}_0, q_n(\hat{\mathbf{S}}(\mathbf{x}^n)) \in \mathcal{B}_{n,\gamma,\zeta}\}$. Lemma 8 bounds success conditional on each such cell by $Cn^k e^{-\zeta\sqrt{n}}$. Consequently,

$$\begin{aligned} \varepsilon_n &\ge P_{XY}^n(\mathcal{B}_n) - P_{XY}^n(\mathcal{A}_n \cap \mathcal{B}_n) \\ &\ge P_{XY}^n(\mathcal{B}_n) - Cn^k e^{-\zeta\sqrt{n}}. \end{aligned} \tag{194}$$

where Eq. (194) averages the conditional bound over disjoint cells; their probabilities sum to at most one.

Recall the safe covariance representative $\mathbf{S}_n := \bar{q}_n(\hat{\mathbf{\Sigma}}_n)$ from (123) and the logarithmic power fluctuations $Z_{j,n}$ from (160). The safe representative makes the converse deficits globally defined. On $\{\hat{\mathbf{\Sigma}}_n \in \mathcal{K}_0\}$,

$$\begin{aligned} A_{j,n}(\mathbf{S}_n) &= \left[\frac{\sqrt{n}}{2}\log\frac{[\mathbf{S}_n]_{jj}}{D_j} - \frac{\log M_{j,n}}{\sqrt{n}}\right]^+ \\ &\ge \left[\frac{\sqrt{n}}{2}\log\frac{[\mathbf{S}_n]_{jj}}{\Sigma_{jj}} - L_j - \gamma\right]^+ \\ &\ge [Z_{j,n} - L_j - \gamma - Cn^{-1/2}]^+, \qquad j \in [1:2]. \end{aligned} \tag{195}$$

where Eq. (195) uses (190), the private-rate envelope, and the uniformly $O(1/n)$ error of quantization inside the logarithm. Coordinatewise monotonicity of ψ preserves the inequality. Localizing the power fluctuations to a bounded set permits (163) and $\rho(\mathbf{S}_n) \to \rho$ to be applied; tightness then removes that localization. The open-set weak-convergence bound at fixed margins and (194) imply

$$\begin{aligned} \liminf_{n\to\infty} \varepsilon_n &\ge \lim_{\gamma\downarrow 0}\lim_{\zeta\downarrow 0} P_{\mathbf{Z}}\{\psi_\rho([Z_1 - L_1 - \gamma]^+, [Z_2 - L_2 - \gamma]^+) > L_0 + \gamma + \zeta\} \\ &= P_{\mathbf{Z}}\{\psi_\rho([Z_1 - L_1]^+, [Z_2 - L_2]^+) > L_0\}. \end{aligned} \tag{196}$$

where Eq. (196) removes margins only after the blocklength limit. The events increase to the strict event by monotonicity and continuity of ψ, so the possible atom at zero causes no difficulty. This matches the two direct bounds and proves case (ii). Restricting to $M_{0,n} = 1$ and $L_0 = 0$ also proves the converse inclusion in (81).

Finally, the displayed regions are closed on the one-sided feasible offset domain. The first is a closed half-space. At the corner, the closed success sets and reverse Fatou give upper semicontinuity of their probabilities in the offsets, so the probability superlevel set is closed. Thus taking the closure in Definition 3 adds no points. Theorem 1 is proved.

VI. Conclusion

We established exact second-order asymptotics for the bivariate Gaussian lossy Gray–Wyner problem. The key direct tool is a covariance-shell covering lemma, which replaces finite-alphabet type covering and yields the scalar normal approximation at every nonzero-common-rate Pareto-minimal boundary tuple. At the zero-common-rate corner, the same local-boundary principle becomes nonlinear and produces a bivariate Gaussian distribution matched by a joint hypercontractive converse. The dispersion at nonzero common rate reduces to the sum of the two supporting slopes, while an absent common message already attains the full $L_0 = 0$ slice.

A discrete corner analogue would require a directional expansion of the type-dependent boundary and a matching joint typewise converse; the differentiable discrete theorem alone does not provide these ingredients. A uniform theorem for operating tuples approaching the corner with blocklength would similarly require joint control of the shrinking first-order common rate and the directional remainder. Several other extensions remain of interest. Gaussian multiterminal problems with decoder side information, including the Heegard–Berger problem [23], [24], require covering and converse arguments that preserve the different observations available to the decoders. Third-order asymptotics would require sharper common-covering and header estimates than the present $o(\sqrt{n})$ remainder, as suggested by point-to-point finite-blocklength refinements [5]. Large- and moderate-deviation analyses would require uniform control on the corresponding shrinking or atypical covariance neighborhoods; the discrete Gray–Wyner and successive-refinement results in [9], [16] provide relevant benchmarks, but the Gaussian corner must retain its nonlinear geometry.

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APPENDIX A
BOUNDARY GEOMETRY AND THE CONVERSE SUPPORT

This appendix first recalls the general-to-Gaussian reduction underlying Proposition 1, then proves Proposition 2 and the information-quantity identities in Section III-B. It also proves Lemma 6 and verifies the fixed-private-rate optimizer used by the code. The common-rate boundary and the entropy relaxation in the imported converse are kept distinct.

A. First-Order Characterization and Boundary Regularity

The unrestricted expected-distortion expression is (22). Its Gaussian evaluation is the existing result of Yu [12, Theorem 3 and Corollary 2], with scaling to the marginal variances in $\mathbf{S}$. To record the connection without repeating that proof, fix $Q \in \mathcal{Q}(Q_{\mathbf{S}}; r_1, r_2)$. Then

$$\begin{aligned} I_Q(X;W) &= I_Q(X;W,\hat{X}) - I_Q(X;\hat{X} \mid W) \\ &\geq I_Q(X;\hat{X}) - r_1 \\ &\geq R_X(Q_{\mathbf{S},X}, D_1) - r_1. \end{aligned} \tag{197}$$

where Eq. (197) uses the chain rule, data processing, the private-rate constraint, and the scalar rate-distortion bound. The same inequality holds for Y. Nonnegativity gives the deficits $\alpha(\mathbf{S}; r_1), \beta(\mathbf{S}; r_2)$, so Yu's unrestricted envelope lower-bounds the common rate by $\Upsilon_{\rho(\mathbf{S})}(\alpha, \beta)$. His attaining Gaussian auxiliary has a constant conditional covariance $\mathbf{K}^\circ \succ 0$ whose marginal information constraints give $K^\circ_{jj} \leq D_j e^{2r_j}$. Hence it is feasible for (27) and attains the same common value.

For clarity about a zero private rate, the standard scalar Gaussian reconstruction can be written explicitly. Let $\tilde{D}_j := \min\{D_j, K_{jj}\}$ and choose independent Gaussian V_j, independent of $(\mathbf{W}, \mathbf{N})$, with variance $\tilde{D}_j(1 - \tilde{D}_j/K_{jj})$. With $\mathbf{X} = \mathbf{W} + \mathbf{N}$ and $\hat{\mathbf{X}} = (\hat{X}, \hat{Y})^{\mathsf{T}}$, set

$$[\hat{\mathbf{X}}]_j := [\mathbf{W}]_j + \left(1 - \frac{\tilde{D}_j}{K_{jj}}\right)[\mathbf{N}]_j + V_j, \tag{198}$$

$$\mathsf{E}_Q[([\mathbf{X}]_j - [\hat{\mathbf{X}}]_j)^2] = \tilde{D}_j \leq D_j, \tag{199}$$

$$I_Q([\mathbf{X}]_j; [\hat{\mathbf{X}}]_j \mid \mathbf{W}) = \tfrac{1}{2}[\log(K_{jj}/D_j)]^+, \qquad j \in [1:2]. \tag{200}$$

where the last two identities follow from independent Gaussian errors; if $D_j \geq K_{jj}$, the reproduction is simply $[\mathbf{W}]_j$. Thus Gaussian substitution gives the reverse bound and

$$\mathsf{R}_{0,\mathrm{E}}(r_1, r_2, D_1, D_2 \mid Q_{\mathbf{S}}) = \mathsf{R}_0(\mathbf{S}; r_1, r_2) = \Upsilon_{\rho(\mathbf{S})}(\alpha(\mathbf{S}; r_1), \beta(\mathbf{S}; r_2)). \tag{201}$$

This is the recalled first-order result, not a deterministic-shell coding statement. The latter is supplied independently by Lemma 3.

We now prove (21), thereby completing the operational statement in Proposition 1. First fix a code sequence satisfying Definition 1. Its encoder can determine the joint successful set $\mathcal{A}_n$ in (14) from the observed block and the known code maps. At blocklength n, add one common symbol. For $m_j \in [1 : M_{j,n}]$, define the modified maps by

$$\tilde{f}(\mathbf{x}^n) := \begin{cases} f(\mathbf{x}^n), & \mathbf{x}^n \in \mathcal{A}_n, \\ (M_{0,n}+1, 1, 1), & \mathbf{x}^n \in \mathcal{E}_n, \end{cases} \tag{202}$$

$$\tilde{\phi}_j(m_0, m_j) := \begin{cases} \phi_j(m_0, m_j), & m_0 \in [1 : M_{0,n}], \\ \mathbf{0}, & m_0 = M_{0,n}+1, \end{cases} \qquad j \in [1:2]. \tag{203}$$

The private alphabets are unchanged, and the common length increases by $\log(M_{0,n}+1) - \log M_{0,n} \leq \log 2$. All new maps are Borel. Write $\tilde{X}^n, \tilde{Y}^n$ for the modified reconstructions. Their first expected distortion obeys

$$\begin{aligned} \mathsf{E}_{P^n_{XY}}[d_X(X^n, \tilde{X}^n)] &\leq D_1 + \mathsf{E}_{P^n_{XY}}[n^{-1} \left\|X^n\right\|^2 \mathbb{1}_{\mathcal{E}_n}] \\ &\leq D_1 + \sqrt{\mathsf{E}_{P^n_{XY}}[(n^{-1}\left\|X^n\right\|^2)^2]\, \varepsilon_n} \\ &= D_1 + \sigma_X^2 \sqrt{(1+2/n)\varepsilon_n} \\ &= D_1 + o(1), \end{aligned} \tag{204}$$

where Eq. (204) uses the zero reconstruction on $\mathcal{E}_n$, Cauchy–Schwarz, and the Gaussian fourth-moment identity. The final term vanishes because $\varepsilon_n \to 0$. The same argument gives a limiting upper bound D_2 for Y. Thus every excess-distortion-achievable tuple belongs to $\mathcal{R}_{\mathrm{E}}$; taking closures proves $\mathcal{R} \subseteq \mathcal{R}_{\mathrm{E}}$.

For the reverse inclusion, fix $(r_1, r_2) \in \mathbb{R}_+^2$ and $\delta > 0$. Use the cell code of Section IV-A with

$$r_0 := \mathsf{R}_0(\mathbf{\Sigma}; r_1, r_2) + \delta. \tag{205}$$

Lemma 4 is available independently of the equivalence being proved: its proof uses only the covariance minimization, deterministic shell covering, and message counting. Its uniform remainder satisfies $h_n/n \to 0$. The safe representative $\mathbf{S}_n$ in (123) converges in probability to $\mathbf{\Sigma}$. Continuity of the covariance value, supplied by Yu's formula and not by an operational identification, gives

$$\begin{aligned}\limsup_{n\to\infty} \varepsilon_n &\le \limsup_{n\to\infty} P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \notin \mathcal{K}_0\} \\ &\quad + \limsup_{n\to\infty} P_{XY}^n\{\mathsf{R}_0(\mathbf{S}_n; r_1, r_2) > \mathsf{R}_0(\mathbf{\Sigma}; r_1, r_2) + \delta\} \\ &= 0,\end{aligned} \tag{206}$$

where Eq. (206) applies the operational shell bound and then the weak law and continuity at $\mathbf{\Sigma}$. The resulting tuple $(\mathsf{R}_0(\mathbf{\Sigma}; r_1, r_2) + \delta, r_1, r_2)$ belongs to $\mathcal{R}$. Sending $\delta \downarrow 0$ after constructing the fixed-margin code sequences and using closure places the covariance graph in $\mathcal{R}$. Equation (201) and upward closure then give $\mathcal{R}_{\mathrm{E}} \subseteq \mathcal{R}$. This proves (21). The same proof applies to every nearby nonsingular Gaussian source under consideration. It does not convert an expected-distortion bound to an excess-distortion bound by Markov's inequality.

The following endpoint reductions explain the special source models mentioned in Section II-A; they do not assert their second-order regions. Let $(\alpha, \beta) \in \mathbb{R}_+^2$ be the private deficits. If $\rho = 0$, independence and the mutual-information chain rule give, for every admissible common auxiliary W with finite information,

$$I_Q(X, Y; W) = I_Q(X; W) + I_Q(Y; W) + I_Q(X; Y \mid W) \tag{207}$$

$$\ge \alpha + \beta. \tag{208}$$

Equation (207) uses $I_Q(X; Y) = 0$, and Eq. (208) uses the two deficit lower bounds established in (197). Independent scalar Gaussian auxiliaries attain equality, with the common message carrying their two descriptions. Thus the first-order requirement is $R_0 \ge \alpha + \beta$.

If $|\rho| = 1$, write $Z := X/\sigma_X = \rho Y/\sigma_Y$. This is a single standard Gaussian source. Each marginal mutual information with a common auxiliary equals $I_Q(Z; W)$, so the common requirement is at least $\max\{\alpha, \beta\}$. Choose a scalar Gaussian common representation with $I_Q(Z; W) = t := \max\{\alpha, \beta\}$ and residual variance e^{-2t}. The two residual source variances are $\Sigma_{jj} e^{-2t}$, giving private rates $[\frac{1}{2}\log(\Sigma_{jj}/D_j) - t]^+ \le r_j$ by (24). This attains the bound, so $R_0 \ge \max\{\alpha, \beta\}$ is the first-order condition. The ambient covariance is singular here; the determinant and Wishart arguments in the main theorem cannot be applied to it.

For the fixed positive-common-rate tuple, define

$$\eta := \partial_\rho \Upsilon_\rho(\alpha^*, \beta^*). \tag{209}$$

In the strict interior of $\mathcal{D}_1$, let $r := r_{\alpha^*, \beta^*}$ and define

$$u := \frac{b_{\beta^*}}{a_{\alpha^*}} e^{\beta^* - \alpha^*}. \tag{210}$$

The derivatives used in the dispersion calculation are

$$\lambda_1^* = \frac{1 - ru}{1 - r^2}, \tag{211}$$

$$\lambda_2^* = \frac{1 - r/u}{1 - r^2}, \tag{212}$$

$$\eta = \frac{r e^{\alpha^* + \beta^*}}{1 - r^2} - \frac{\rho}{1 - \rho^2}. \tag{213}$$

On the affine branches and their adjoining interfaces,

$$(\lambda_1^*, \lambda_2^*, \eta) = \begin{cases} (1, 1, -\rho/(1 - \rho^2)), & \text{if } (\alpha^*, \beta^*) \in \mathcal{D}_2, \\ (1, 0, 0), & \text{if } (\alpha^*, \beta^*) \in \mathcal{D}_3, \\ (0, 1, 0), & \text{if } (\alpha^*, \beta^*) \in \mathcal{D}_4. \end{cases} \tag{214}$$

We verify these formulas and their agreement at the interfaces as part of the regularity proof.

We now verify the regularity in Proposition 2. For positive (α, β), abbreviate $a = a_\alpha$, $b = b_\beta$, $r = r_{\alpha,\beta}$, and $u = (b/a)e^{\beta-\alpha}$. Let F_k be the kth expression in (40). Differentiation of the curved branch gives

$$\begin{aligned} \partial_\alpha r &= r - u, \qquad (215)\\ \partial_\beta r &= r - u^{-1},\\ \partial_\rho r &= e^{\alpha+\beta},\\ \partial_\alpha F_1 &= 1 + \frac{r\partial_\alpha r}{1-r^2} = \frac{1-ru}{1-r^2}, \qquad (216)\\ \partial_\beta F_1 &= \frac{1-r/u}{1-r^2},\\ \partial_\rho F_1 &= \frac{re^{\alpha+\beta}}{1-r^2} - \frac{\rho}{1-\rho^2}. \qquad (217) \end{aligned}$$

where Eqs. (215)–(217) follow from $a' = e^{-2\alpha}/a$, $b' = e^{-2\beta}/b$, and the chain rule. In the strict curved interior, $r > 0$, $r - u = e^{\alpha+\beta}(\rho - b/a) < 0$, and $r - u^{-1} = e^{\alpha+\beta}(\rho - a/b) < 0$. Hence $0 < r < \min\{u, u^{-1}\}$, which gives $0 < \lambda_j^* < 1$.

With gradients in (α, β, ρ), substitution at the three interfaces yields

$$\begin{aligned} \nabla F_1|_{r=0} &= (1, 1, -\rho/(1-\rho^2)) = \nabla F_2, \qquad (218)\\ \nabla F_1|_{\rho=b/a} &= (1, 0, 0) = \nabla F_3, \qquad (219)\\ \nabla F_1|_{\rho=a/b} &= (0, 1, 0) = \nabla F_4. \qquad (220) \end{aligned}$$

where the second substitution uses $r = u$ and the third is its coordinate interchange. The branch values agree as well. The finitely many adjoining analytic pieces therefore have a common first-order expansion and continuous first derivatives.

Near $\alpha = 0$, $\beta > 0$, the active expression is $F_4 = \beta$, independent of α; the other positive axis is symmetric. This handles the positive-part composition. If $R_j^* = 0$, its deficit is positive because $D_j < \Sigma_{jj}$, so the same branch expressions extend to small negative r_j for differentiation. Operational rates remain nonnegative. Only the deficit origin is excluded. This proves the stated continuously differentiable extension, including all finite interfaces and zero private rates.

For the zero-slope interpretation in Section III-B, fix a positive interface pair (α, β) with $b_\beta = \rho a_\alpha$. Keep α fixed and let $h > 0$ increase the second deficit into $\mathcal{D}_1$. At the interface $r = u$, while $\partial_\beta r = r - u^{-1}$ and $\partial_\beta \log u = b_\beta^{-2}$. Differentiating the curved branch from the right gives

$$\begin{aligned} \partial_{\beta,+}^2 \Upsilon_\rho(\alpha, \beta) &= u^{-2} + \frac{1}{b_\beta^2(1-u^2)}\\ &= \frac{(1-\rho^2 a_\alpha^2)^2}{\rho^2 a_\alpha^2(1-a_\alpha^2)(1-\rho^2)} > 0, \qquad (221)\\ \Upsilon_\rho(\alpha, \beta+h) &= \alpha + \frac{(1-\rho^2 a_\alpha^2)^2}{2\rho^2 a_\alpha^2(1-a_\alpha^2)(1-\rho^2)} h^2 + o(h^2), \qquad h \downarrow 0. \qquad (222) \end{aligned}$$

Here Eq. (221) substitutes $b_\beta^2 = \rho^2 a_\alpha^2$ and $u^2 = \rho^2(1-a_\alpha^2)/(1-\rho^2 a_\alpha^2)$; Eq. (222) uses the zero first derivative. For sufficiently small negative h, the value remains α on $\mathcal{D}_3$. This expansion fixes a positive interface pair and is not asserted uniformly as that pair approaches the corner.

B. Covariance Influence, Dispersion, and Converse Support

For $Q_{\mathbf{S}} = \mathcal{N}(\mathbf{0}, \mathbf{S})$, the mean of a quadratic form and Proposition 2 give

$$\begin{aligned} \mathsf{E}_{Q_{\mathbf{S}}}[\jmath_{XY}(X, Y)] &= R_0^* + \operatorname{tr}(\mathbf{G}\mathbf{S}) - \operatorname{tr}(\mathbf{G}\mathbf{\Sigma}) = R_0^* + \langle \mathbf{G}, \mathbf{S} - \mathbf{\Sigma} \rangle, \qquad (223)\\ \mathsf{R}_0(\mathbf{S}; R_1^*, R_2^*) &= \mathsf{E}_{Q_{\mathbf{S}}}[\jmath_{XY}(X, Y)] + o(\|\mathbf{S} - \mathbf{\Sigma}\|). \qquad (224) \end{aligned}$$

where Eq. (223) uses the definition of $\jmath_{XY}$ and Eq. (224) is the boundary expansion at fixed private rates. The mean identity follows by setting $\mathbf{S} = \mathbf{\Sigma}$.

For a symmetric perturbation $d\mathbf{S}$ at $\mathbf{\Sigma}$, the chain rule gives

$$\begin{aligned} d\rho(\mathbf{S}) &= \frac{dS_{12}}{\sigma_X \sigma_Y} - \frac{\rho}{2\sigma_X^2} dS_{11} - \frac{\rho}{2\sigma_Y^2} dS_{22}, \qquad (225)\\ d\mathsf{R}_0 &= \lambda_1^* \frac{dS_{11}}{2\sigma_X^2} + \lambda_2^* \frac{dS_{22}}{2\sigma_Y^2} + \eta\, d\rho(\mathbf{S}) = \langle \mathbf{G}, d\mathbf{S} \rangle. \qquad (226) \end{aligned}$$

where Eq. (226) uses the marginal logarithmic derivatives and counts the symmetric off-diagonal entry twice, giving (227). At a positive axis the vanishing deficit has zero boundary derivative, as proved above.

Differentiating (28) and (29) gives the explicit matrix

$$\mathbf{G} = \frac{1}{2}\begin{bmatrix} (\lambda_1^* - \rho\eta)/\sigma_X^2 & \eta/(\sigma_X\sigma_Y) \\ \eta/(\sigma_X\sigma_Y) & (\lambda_2^* - \rho\eta)/\sigma_Y^2 \end{bmatrix}. \tag{227}$$

Let $\mathbf{\Omega} = \mathbf{\Sigma}^{1/2}\mathbf{G}\mathbf{\Sigma}^{1/2}$ have eigenvalues ω_1, ω_2, and let N_1, N_2 be independent standard Gaussians. Orthogonal diagonalization yields

$$\jmath_{XY}(X,Y) - R_0^* \overset{d}{=} \omega_1(N_1^2 - 1) + \omega_2(N_2^2 - 1), \tag{228}$$

$$\mathsf{Var}_{P_{XY}}[\jmath_{XY}(X,Y)] = 2(\omega_1^2 + \omega_2^2) = 2\,\mathrm{tr}(\mathbf{\Omega}^2) = 2\,\mathrm{tr}[(\mathbf{G}\mathbf{\Sigma})^2]. \tag{229}$$

where Eq. (229) uses independence and $\mathsf{Var}(N_j^2) = 2$. This representation also gives finite third absolute moment. The following argument determines the eigenvalues, making a second expanded variance formula unnecessary.

To evaluate the dispersion through its spectrum, fix a Pareto-minimal boundary tuple $\mathbf{R}^*$ with $R_0^* > 0$, the weights λ_1^*, λ_2^* from (54)–(55), and the derivative $\mathbf{G}$ from (65). Let $\mathbf{K}^*$ be the optimizer of (27) at $(\mathbf{\Sigma}, R_1^*, R_2^*)$, whose existence is proved in Lemma 2. Define the positive caps $\tau_j := D_j e^{2R_j^*}$ for $j \in [1:2]$.

After dropping the constant $\frac{1}{2}\log|\mathbf{\Sigma}|$, the primal problem minimizes $-\frac{1}{2}\log|\mathbf{K}|$ under the affine diagonal caps and covariance order. A sufficiently small positive multiple of $\mathbf{I}_2$ is strictly feasible. Convex duality therefore gives multipliers $\vartheta_1, \vartheta_2 \geq 0$ and $\mathbf{\Lambda} \in \mathbb{S}_+^2$ satisfying

$$-\tfrac{1}{2}(\mathbf{K}^*)^{-1} + \mathrm{diag}(\vartheta_1, \vartheta_2) + \mathbf{\Lambda} = \mathbf{0}, \tag{230}$$

$$\langle \mathbf{\Lambda}, \mathbf{\Sigma} - \mathbf{K}^* \rangle = 0. \tag{231}$$

For a nearby positive-definite $\mathbf{S}$, the same dual multipliers are feasible. Their dual objective changes only by $-\langle \mathbf{\Lambda}, \mathbf{S} - \mathbf{\Sigma} \rangle$. Consequently,

$$\mathsf{R}_0(\mathbf{S}; R_1^*, R_2^*) - \tfrac{1}{2}\log|\mathbf{S}| \geq R_0^* - \tfrac{1}{2}\log|\mathbf{\Sigma}| - \langle \mathbf{\Lambda}, \mathbf{S} - \mathbf{\Sigma} \rangle, \tag{232}$$

$$\mathbf{G} = \tfrac{1}{2}\mathbf{\Sigma}^{-1} - \mathbf{\Lambda}, \tag{233}$$

where Eq. (232) is weak duality with equality at $\mathbf{S} = \mathbf{\Sigma}$, and Eq. (233) follows by differentiating the touching inequality in both signs of every symmetric direction, using Proposition 2. No differentiability of the optimizer or of its active set is needed.

Let $\mathbf{\Delta}_* := \mathbf{\Sigma} - \mathbf{K}^* \succeq 0$. Since $R_0^* > 0$, the covariance $\mathbf{K}^*$ cannot equal $\mathbf{\Sigma}$, so $\mathbf{\Delta}_* \neq \mathbf{0}$. For two positive-semidefinite matrices, zero trace product implies zero product: indeed $\mathrm{tr}(\mathbf{\Lambda}\mathbf{\Delta}_*) = \left\|\mathbf{\Lambda}^{1/2}\mathbf{\Delta}_*^{1/2}\right\|^2 = 0$, so $\mathbf{\Lambda}^{1/2}\mathbf{\Delta}_*^{1/2} = \mathbf{0}$ and then $\mathbf{\Lambda}\mathbf{\Delta}_* = \mathbf{0}$. Equation (231) therefore gives

$$\mathbf{\Lambda}\mathbf{\Delta}_* = \mathbf{0}, \qquad \mathrm{rank}(\mathbf{\Lambda}) \leq 1, \tag{234}$$

$$\mathbf{\Omega} := \mathbf{\Sigma}^{1/2}\mathbf{G}\mathbf{\Sigma}^{1/2} = \tfrac{1}{2}\mathbf{I}_2 - \mathbf{\Sigma}^{1/2}\mathbf{\Lambda}\mathbf{\Sigma}^{1/2}, \tag{235}$$

$$\mathrm{tr}(\mathbf{\Omega}) = \mathrm{tr}(\mathbf{G}\mathbf{\Sigma}) = \frac{\lambda_1^* + \lambda_2^*}{2}, \tag{236}$$

where Eq. (234) uses the nonzero range of $\mathbf{\Delta}_*$, Eq. (235) substitutes (233), and Eq. (236) evaluates (227). The rank bound forces one eigenvalue of $\mathbf{\Omega}$ to equal $1/2$; the trace determines the other as $(\lambda_1^* + \lambda_2^* - 1)/2$.

For completeness, the slope sum lies in $[1, 2]$. The upper bound follows from $\lambda_j^* \leq 1$. In the strict curved region, the parameters r, u in (210)–(212) satisfy

$$\begin{aligned} \lambda_1^* + \lambda_2^* - 1 &= \frac{1 + r^2 - r(u + u^{-1})}{1 - r^2} \\ &= \frac{(u - r)(1 - ru)}{u(1 - r^2)} > 0, \end{aligned} \tag{237}$$

where Eq. (237) follows by substitution and factorization, and positivity uses $0 < r < \min\{u, u^{-1}\}$. The affine values and all finite interfaces follow from (214) and continuity. Thus

$$\begin{aligned} \mathsf{Var}_{P_{XY}}[\jmath_{XY}(X,Y)] &= 2\,\mathrm{tr}(\mathbf{\Omega}^2) \\ &= 2\left[\left(\frac{1}{2}\right)^2 + \left(\frac{\lambda_1^* + \lambda_2^* - 1}{2}\right)^2\right] \\ &= \frac{1 + (\lambda_1^* + \lambda_2^* - 1)^2}{2} = V, \end{aligned} \tag{238}$$

where Eq. (238) uses the two identified eigenvalues. This proves the variance identification in (69) and the bounds (57)–(58), including zero private rates, because the diagonal caps remain strictly positive and Slater's condition remains valid there.

The same diagonalization identifies the exact finite-blocklength distribution used in Section III-C. Let $\Xi_{1,n}, \Xi_{2,n}$ be independent chi-squared variables with n degrees of freedom. Independence across source letters and the two whitened coordinates gives

$$\begin{aligned}\frac{1}{\sqrt{n}} \sum_{i\in[1:n]} [\jmath_{XY}(X_i, Y_i) - R_0^*] &\overset{d}{=} \frac{1}{2\sqrt{n}} \sum_{i\in[1:n]} (N_{1,i}^2 - 1) + \frac{\lambda_1^* + \lambda_2^* - 1}{2\sqrt{n}} \sum_{i\in[1:n]} (N_{2,i}^2 - 1) \\ &\overset{d}{=} \frac{\Xi_{1,n} - n}{2\sqrt{n}} + \frac{\lambda_1^* + \lambda_2^* - 1}{2\sqrt{n}} (\Xi_{2,n} - n), \end{aligned} \tag{239}$$

where Eq. (239) sums independent squared standard Gaussian variables $N_{j,i}$. This is an exact distributional identity for the linearized source statistic, not for the error probability of the covering code.

We prove Lemma 6. The positive weights are fixed. Time sharing makes Yu's unrestricted lower envelope convex. For $\mathbf{0} \prec \mathbf{K} \preceq \mathbf{\Sigma}$, let $\alpha_{\mathbf{K}} = \frac{1}{2}\log(\sigma_X^2/K_{11})$, $\beta_{\mathbf{K}} = \frac{1}{2}\log(\sigma_Y^2/K_{22})$, and $t_{\mathbf{K}} = \frac{1}{2}\log(|\mathbf{\Sigma}|/|\mathbf{K}|)$. The supporting plane at (α^*, β^*) gives

$$\begin{aligned} t_{\mathbf{K}} &\geq \Upsilon_\rho(\alpha_{\mathbf{K}}, \beta_{\mathbf{K}}) \\ &\geq R_0^* + \lambda_1^*(\alpha_{\mathbf{K}} - \alpha^*) + \lambda_2^*(\beta_{\mathbf{K}} - \beta^*), \end{aligned} \tag{240}$$

$$\begin{aligned} \frac{1}{2}\log\frac{|\mathbf{\Sigma}|}{|\mathbf{K}|} + \sum_{j\in[1:2]} \frac{\lambda_j^*}{2}\log\frac{K_{jj}}{D_j} &= t_{\mathbf{K}} - \lambda_1^*\alpha_{\mathbf{K}} - \lambda_2^*\beta_{\mathbf{K}} + \sum_{j\in[1:2]} \frac{\lambda_j^*}{2}\log\frac{\Sigma_{jj}}{D_j} \\ &\geq R_0^* + \lambda_1^* R_1^* + \lambda_2^* R_2^*. \end{aligned} \tag{241}$$

where Eq. (240) uses the unrestricted envelope and convexity, and Eq. (241) substitutes the boundary deficits. At a common-rate optimizer $\mathbf{K}^*$, feasibility gives $\alpha_{\mathbf{K}^*} \geq \alpha^*$ and $\beta_{\mathbf{K}^*} \geq \beta^*$. Since $t_{\mathbf{K}^*} = R_0^*$ and both weights are positive, Eq. (240) forces both inequalities to be equalities. Thus $\mathbf{K}^*$ attains the last bound, proving (177), including $R_j^* = 0$ where $K_{jj}^* = D_j$.

For nearby $\mathbf{S}$, evaluating the entropy support at a common-rate optimizer $\mathbf{K}_{\mathbf{S}}^*$ yields

$$\begin{aligned} \underline{F}_{\boldsymbol{\lambda}^*}(\mathbf{S}) &\leq \frac{1}{2}\log\frac{|\mathbf{S}|}{|\mathbf{K}_{\mathbf{S}}^*|} + \sum_{j\in[1:2]} \frac{\lambda_j^*}{2}\log\frac{[\mathbf{K}_{\mathbf{S}}^*]_{jj}}{D_j} \\ &\leq \mathsf{R}_0(\mathbf{S}; R_1^*, R_2^*) + \lambda_1^* R_1^* + \lambda_2^* R_2^*. \end{aligned} \tag{242}$$

where Eq. (242) uses the private covariance caps. Equality holds at $\mathbf{\Sigma}$, not necessarily on a neighborhood.

Recall the concave function $\varphi_{\boldsymbol{\lambda}^*,\mathbf{\Sigma}}$, its matrix supergradient $\mathbf{B}_{\boldsymbol{\lambda}^*}$, and centered quadratic function $g_{\boldsymbol{\lambda}^*}$ defined in Section V-A. The covariance/functional-supergradient correspondence is the imported result [11, Proposition 5]; we only identify its matrix here. Since $\varphi_{\boldsymbol{\lambda}^*,\mathbf{\Sigma}}$ is finite and concave on the open positive-definite cone, it has a supergradient at $\mathbf{\Sigma}$. Equation (242) provides a differentiable lower touching function for $\varphi_{\boldsymbol{\lambda}^*,\mathbf{\Sigma}}$. Its derivative at $\mathbf{\Sigma}$ is $-\mathbf{G}$, since the derivatives of the additional log-determinant and reference-linear terms cancel. For $\mathbf{H} \in \mathbb{S}^2$ and sufficiently small $t > 0$,

$$\begin{aligned} -\langle \mathbf{G}, \mathbf{H}\rangle + o(1) &\leq \frac{\varphi_{\boldsymbol{\lambda}^*,\mathbf{\Sigma}}(\mathbf{\Sigma} + t\mathbf{H}) - \varphi_{\boldsymbol{\lambda}^*,\mathbf{\Sigma}}(\mathbf{\Sigma})}{t} \\ &\leq \langle \mathbf{B}_{\boldsymbol{\lambda}^*}, \mathbf{H}\rangle . \end{aligned} \tag{243}$$

where Eq. (243) combines the touching bound with (175). Sending $t \downarrow 0$ in the directions $\mathbf{H}$ and $-\mathbf{H}$ gives

$$\langle \mathbf{B}_{\boldsymbol{\lambda}^*}, \mathbf{H}\rangle = -\langle \mathbf{G}, \mathbf{H}\rangle , \qquad \mathbf{H} \in \mathbb{S}^2, \tag{244}$$

$$g_{\boldsymbol{\lambda}^*}(\mathbf{x}) = -[\jmath_{XY}(x, y) - R_0^*]. \tag{245}$$

where the second identity centers the quadratic form with matrix $-\mathbf{G}$. The unique supergradient makes $\varphi_{\boldsymbol{\lambda}^*,\mathbf{\Sigma}}$ differentiable at this interior point, and (174) then gives (178). The variance is V by (69). This establishes all parts of Lemma 6 without differentiating a clipped private logarithm.

We complete the covariance calculations by evaluating and verifying the selected optimizer. Fix a covariance $\mathbf{S} = [s_{jk}] \in \mathcal{K}$ and private rates $(r_1, r_2) \in \mathbb{R}_+^2$. Write $s_j := s_{jj}$ for $j \in [1:2]$, $\rho_{\mathbf{S}} := \rho(\mathbf{S})$, $\alpha := \alpha(\mathbf{S}; r_1)$, and $\beta := \beta(\mathbf{S}; r_2)$. Define

$$\mathbf{T}_{\mathbf{S}} := \operatorname{diag}(\sqrt{s_1}, \sqrt{s_2}), \tag{246}$$

$$v_1 := e^{-2\alpha} = \min\{1, D_1 e^{2r_1}/s_1\}, \tag{247}$$

$$v_2 := e^{-2\beta} = \min\{1, D_2 e^{2r_2}/s_2\}. \tag{248}$$

Here $a_\alpha = \sqrt{1 - v_1}$ and $b_\beta = \sqrt{1 - v_2}$ retain their definitions in (33)–(34). Let $\tilde{\mathbf{K}}^* \in \mathbb{S}_{++}^2$ be the normalized residual matrix

specified below. The unique optimizer of the fixed-private-rate problem (27) is

$$\mathbf{K}^*(\mathbf{S}; r_1, r_2) = \mathbf{T}_{\mathbf{S}} \tilde{\mathbf{K}}^* \mathbf{T}_{\mathbf{S}}, \tag{249}$$

where the branches below use correlation $\rho_{\mathbf{S}}$:

$$\tilde{\mathbf{K}}^* = \begin{cases} \begin{bmatrix} v_1 & \rho_{\mathbf{S}} - a_\alpha b_\beta \\ \rho_{\mathbf{S}} - a_\alpha b_\beta & v_2 \end{bmatrix}, & (\alpha, \beta) \in \mathcal{D}_1, \\ \begin{bmatrix} v_1 & 0 \\ 0 & v_2 \end{bmatrix}, & (\alpha, \beta) \in \mathcal{D}_2, \\ \begin{bmatrix} v_1 & \rho_{\mathbf{S}} v_1 \\ \rho_{\mathbf{S}} v_1 & 1 - \rho_{\mathbf{S}}^2 + \rho_{\mathbf{S}}^2 v_1 \end{bmatrix}, & (\alpha, \beta) \in \mathcal{D}_3, \\ \begin{bmatrix} 1 - \rho_{\mathbf{S}}^2 + \rho_{\mathbf{S}}^2 v_2 & \rho_{\mathbf{S}} v_2 \\ \rho_{\mathbf{S}} v_2 & v_2 \end{bmatrix}, & (\alpha, \beta) \in \mathcal{D}_4. \end{cases} \tag{250}$$

The expressions agree on shared interfaces and extend continuously to the axes; at $(\alpha, \beta) = (0, 0)$, take $\mathbf{K}^* = \mathbf{S}$. To verify the formulas, let $\tilde{\mathbf{S}} := \mathbf{T}_{\mathbf{S}}^{-1} \mathbf{S} \mathbf{T}_{\mathbf{S}}^{-1}$. For the candidates in (250), the covariance differences on $\mathcal{D}_1$ and $\mathcal{D}_3$ are

$$\tilde{\mathbf{S}} - \tilde{\mathbf{K}}^* = \begin{bmatrix} a_\alpha \\ b_\beta \end{bmatrix} \begin{bmatrix} a_\alpha & b_\beta \end{bmatrix} \succeq 0, \qquad (\alpha, \beta) \in \mathcal{D}_1, \tag{251}$$

$$\tilde{\mathbf{S}} - \tilde{\mathbf{K}}^* = (1 - v_1) \begin{bmatrix} 1 \\ \rho_{\mathbf{S}} \end{bmatrix} \begin{bmatrix} 1 & \rho_{\mathbf{S}} \end{bmatrix} \succeq 0, \qquad (\alpha, \beta) \in \mathcal{D}_3. \tag{252}$$

where both identities follow by subtraction. On $\mathcal{D}_2$, the difference has nonnegative diagonal entries and determinant $a_\alpha^2 b_\beta^2 - \rho_{\mathbf{S}}^2 \geq 0$. On $\mathcal{D}_3$, its second private cap follows from $1 - \rho_{\mathbf{S}}^2(1 - v_1) \leq v_2$; the fourth branch is symmetric. The residual determinants are

$$|\tilde{\mathbf{K}}^*| = \begin{cases} v_1 v_2 (1 - r_{\alpha,\beta}^2), & (\alpha, \beta) \in \mathcal{D}_1, \\ v_1 v_2, & (\alpha, \beta) \in \mathcal{D}_2, \\ v_1 (1 - \rho_{\mathbf{S}}^2), & (\alpha, \beta) \in \mathcal{D}_3, \\ v_2 (1 - \rho_{\mathbf{S}}^2), & (\alpha, \beta) \in \mathcal{D}_4. \end{cases} \tag{253}$$

These are positive, verifying positive definiteness and all caps. Finally,

$$\begin{aligned} \frac{1}{2} \log \frac{|\mathbf{S}|}{|\mathbf{K}^*|} &= \frac{1}{2} \log \frac{1 - \rho_{\mathbf{S}}^2}{|\tilde{\mathbf{K}}^*|} \\ &= \Upsilon_{\rho_{\mathbf{S}}}(\alpha, \beta) = \mathsf{R}_0(\mathbf{S}; r_1, r_2), \end{aligned} \tag{254}$$

where Eq. (254) uses determinant scaling, (253), and Yu's formula. Strict concavity of $\log |\mathbf{K}|$ on the convex feasible set proves uniqueness. The candidates agree on interfaces and extend to the axes; at zero deficits, $\mathbf{K}^* = \mathbf{S}$.

Appendix B
The Uniform Corner Expansion

We prove Lemma 5, including the moving branch interfaces and bounded correlation perturbations.

For $a, b \geq 0$, monotonicity in the hypercontractive exponents places the optimization on $(p - 1)(q - 1) = \rho^2$. With $s = p - 1 > 0$, its objective and derivative are

$$\psi_\rho(a, b) = \sup_{s > 0} \left\{ \frac{a}{1 + s} + \frac{bs}{s + \rho^2} \right\}, \tag{255}$$

$$\frac{d}{ds} \left(\frac{a}{1 + s} + \frac{bs}{s + \rho^2} \right) = -\frac{a}{(1 + s)^2} + \frac{b\rho^2}{(s + \rho^2)^2}. \tag{256}$$

where Eq. (255) substitutes $q - 1 = \rho^2 / s$. For positive a, b, the stationary point is $s = (\rho\sqrt{b} - \rho^2\sqrt{a})/(\sqrt{a} - \rho\sqrt{b})$, positive exactly when $\rho^2 a < b < a/\rho^2$. The derivative changes from positive to negative there; substitution gives the curved expression in (61). Otherwise the supremum is a as $s \downarrow 0$ or b as $s \uparrow \infty$. These limits also give the axes and interfaces.

The compact form of the same optimization is

$$\mathcal{C}_\rho := \{(u,v) \in [0,1]^2 : (1-u)(1-v) \geq \rho^2 uv\}, \tag{257}$$

$$\psi_\rho(a,b) = \max_{(u,v)\in\mathcal{C}_\rho} (ua + vb), \tag{258}$$

$$\max\{a,b\} \leq \psi_\rho(a,b) \leq a+b, \tag{259}$$

$$\begin{aligned} |\psi_\rho(a,b) - \psi_\rho(a',b')| &\leq \max_{(u,v)\in\mathcal{C}_\rho} |u(a-a') + v(b-b')| \\ &\leq |a-a'| + |b-b'|. \end{aligned} \tag{260}$$

where Eq. (258) takes the closure of $(1/p, 1/q)$, Eq. (259) uses $(1,0),(0,1) \in \mathcal{C}_\rho \subset [0,1]^2$, and Eq. (260) compares the two maxima. This also proves positive homogeneity and coordinatewise monotonicity. On bounded argument sets and compact subintervals of $\rho \in (0,1)$, the curved formula has a bounded ρ derivative; the constant one-source pieces join continuously. Thus ψ_ρ is locally Lipschitz in ρ, uniformly on those sets.

To prove the uniform expansion, fix $0 \leq a, b \leq M$. Set $A_t(a) = (1-e^{-2ta})/(2t)$ for $t > 0$. Uniformly on this square,

$$A_t(a) = a\int_0^1 e^{-2tau}\,du = a + O_M(t), \tag{261}$$

$$\sqrt{(1-e^{-2ta})(1-e^{-2tb})} = 2t\sqrt{ab} + O_M(t^2), \tag{262}$$

$$r_{ta,tb} = \rho + t[\rho(a+b) - 2\sqrt{ab}] + O_M(t^2). \tag{263}$$

where Eq. (262) expands $\sqrt{(1-e^{-2x})/(2x)}$ continuously at zero, and Eq. (263) then expands the exponential in (35). No division by a vanishing argument occurs. The product in (262) is smaller than ρ for small t, so $\mathcal{D}_2$ is excluded uniformly. Write $\psi_1(a,b) = (a+b-2\rho\sqrt{ab})/(1-\rho^2)$. On the finite-$t$ curved branch,

$$\begin{aligned} \Upsilon_\rho(ta,tb) &= t(a+b) - \frac{1}{2}\log\frac{1-r_{ta,tb}^2}{1-\rho^2} \\ &= t\psi_1(a,b) + O_M(t^2), \end{aligned} \tag{264}$$

where Eq. (264) applies the logarithmic Taylor expansion to (263); $r_{ta,tb}$ remains uniformly bounded away from one, so the logarithm's argument remains bounded away from zero. The finite-t one-source expressions are exactly ta and tb.

The exact interface with $\mathcal{D}_3$ is $A_t(b) = \rho^2 A_t(a)$. A disagreement with the limiting classifier $b = \rho^2 a$ can occur only in the strip

$$|b - \rho^2 a| \leq |b - A_t(b)| + \rho^2|a - A_t(a)| \leq C_M t, \tag{265}$$

$$0 \leq \psi_1(a,b) - a = \frac{(\sqrt{b} - \rho\sqrt{a})^2}{1-\rho^2} \leq \frac{|b-\rho^2 a|}{1-\rho^2}. \tag{266}$$

where Eq. (265) uses the opposite signs of the two classifiers and (261), and Eq. (266) uses $(\sqrt{x}-\sqrt{y})^2 \leq |x-y|$. Interchanging a, b gives the other strip and gap. If both one-source classifiers switch, both estimates apply. Thus the coefficient error is $O_M(t)$ wherever the branches disagree, and

$$\sup_{0\leq a,b\leq M} |\Upsilon_\rho(ta,tb) - t\psi_\rho(a,b)| \leq C_M t^2 + tC_M t \leq C'_M t^2. \tag{267}$$

where Eq. (267) combines the fixed-branch remainder with the strip bounds, including the origin.

For $\rho_t = \rho + th$, $|h| \leq H$, all correlations lie in one compact subinterval of $(0,1)$ for small t. The same constants can therefore be used, and

$$\begin{aligned} |\Upsilon_{\rho_t}(ta,tb) - t\psi_\rho(a,b)| &\leq |\Upsilon_{\rho_t}(ta,tb) - t\psi_{\rho_t}(a,b)| + t|\psi_{\rho_t}(a,b) - \psi_\rho(a,b)| \\ &\leq C_M t^2 + tC_{M,H} t. \end{aligned} \tag{268}$$

where Eq. (268) uses (267) uniformly in correlation and the Lipschitz property proved above. This is the uniform estimate in Lemma 5.

For a positive level ℓ, homogeneity and (259) make each ray in $(0,\infty)^2$ meet $\{\psi_\rho = \ell\}$ in a single point. Polar integration gives this level set zero area. Where one positive-part argument vanishes, its preimage at a positive level is a horizontal or vertical line. Since $P_{\mathbf{Z}}$ has a density, $T(\mathbf{Z})$ has no positive atom. Finally, (259) gives $\psi_\rho(a,b) = 0$ exactly at the origin, proving (148).

APPENDIX C
PROOFS OF THE CODING LEMMAS

Fix the distortions, shell width, and compact covariance bounds in Lemma 3. All constants in this appendix may depend on those fixed quantities, but not on n, the selected pair $(\mathbf{S}, \mathbf{K})$, or the sequence in its shell. We analyze the code already defined in Section IV-A. The common-codebook distribution is $Q_{0,n}$ in (110), and the private codebooks and decoding functions are fixed there. The auxiliary posterior below is an analysis distribution, not an extra random operation performed by the encoder. Probabilities under P_{XY} refer only to the actual source.

A. Euclidean Covering and Gaussian Preparations

We prove Lemma 1. Fix $T \geq 1$, $n \geq 3$, $r > 0$, and $0 \leq R \leq Tr$. If $R \leq r$, the origin suffices. Suppose $R > r$, and define

$$h := \frac{r}{4n}, \qquad r_- := r - 2h, \tag{269}$$

$$m := \left\lceil \frac{R-r}{h} \right\rceil, \qquad R_k := \min\{r + kh, R\}, \quad k \in [1:m]. \tag{270}$$

The radii satisfy $r < R_k \leq R$. Every point with radius in $(r, R]$ is within distance h of a point with the same direction and one of the radii R_k. The inner ball of radius r will be covered by the origin.

For each sphere $\mathcal{S}_{n,k} := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| = R_k\}$, choose an h-net $\mathcal{N}_{n,k} \subset \mathcal{S}_{n,k}$ by taking a maximal h-separated set. The disjoint radius-$h/2$ balls around its points give

$$\begin{aligned} |\mathcal{N}_{n,k}| &\leq \left(1 + \frac{2R_k}{h}\right)^n \\ &\leq (1 + 8Tn)^n, \end{aligned} \tag{271}$$

where Eq. (271) is the ambient-volume comparison and $R_k/r \leq T$.

Choose a center uniformly on the sphere of radius $\sqrt{R_k^2 - r_-^2}$. For a fixed net point, the radius-r_- ball about this center covers it when the cosine of the angle is at least $z_k := \sqrt{1 - r_-^2/R_k^2}$. Write $p_{n,k}$ for this probability and $\omega_n := \Gamma(n/2)/(\sqrt{\pi}\,\Gamma((n-1)/2))$. The coordinate density of a uniform sphere point gives

$$\begin{aligned} p_{n,k} &= \omega_n \int_{z_k}^1 (1-u^2)^{(n-3)/2}\, du \\ &= \frac{\omega_n}{2} \int_0^{1-z_k^2} w^{(n-3)/2}(1-w)^{-1/2}\, dw \\ &\geq \frac{1}{2(n-1)} \left(\frac{r_-}{R_k}\right)^{n-1}, \end{aligned} \tag{272}$$

where Eq. (272) uses the spherical-coordinate density, the substitution $w = 1 - u^2$, $\omega_n \geq 1/2$ for $n \geq 3$, and $(1-w)^{-1/2} \geq 1$. Importantly, this lower bound remains valid as $z_k \downarrow 0$; no positive lower bound on the cap cutoff is used.

Take

$$J_{n,k} := \left\lceil 2(n-1)\left(\frac{R_k}{r_-}\right)^{n-1} \left[n \log(1 + 8Tn) + 1\right] \right\rceil \tag{273}$$

independent centers of that radius. Let $\mathcal{F}_{n,k}$ be the event that at least one point of $\mathcal{N}_{n,k}$ is uncovered by their radius-r_- balls. Then

$$\begin{aligned} \Pr(\mathcal{F}_{n,k}) &\leq |\mathcal{N}_{n,k}|(1 - p_{n,k})^{J_{n,k}} \\ &\leq |\mathcal{N}_{n,k}| e^{-J_{n,k} p_{n,k}} \\ &\leq e^{-1} < 1, \end{aligned} \tag{274}$$

where Eq. (274) uses the union bound, independence, $1 - x \leq e^{-x}$, and (271)–(273). Fix a covering realization for each of the finitely many radii. For an arbitrary point of the outer ball, radial rounding costs at most h and the sphere net costs at most another h. Its distance to a selected center is therefore at most $r_- + 2h = r$. Together with the origin, these centers cover $\mathcal{B}_n(\mathbf{0}, R)$.

Finally, $m \le 4nT+1$, and

$$\left(\frac{r}{r_-}\right)^{n-1} = \left(1-\frac{1}{2n}\right)^{-(n-1)} \le \exp\left(\frac{n-1}{2n-1}\right) \le e^{1/2}, \tag{275}$$

where Eq. (275) uses $-\log(1-x) \le x/(1-x)$. Since $R_k \le R$ and $R/r > 1$, the total number of centers obeys

$$\begin{aligned} |\mathcal{V}| &\le 1 + \sum_{k=1}^{m} J_{n,k} \\ &\le C_T n^3 \big[1+\log n\big](R/r)^n \\ &\le C'_T n^4 (R/r)^n, \end{aligned} \tag{276}$$

where Eq. (276) uses (273), $m \le 4nT+1$, (275), and $\log(1+8Tn) \le \log(1+8T) + \log n$. The constants depend only on T, and the proof works for every $n \ge 3$. Absorbing C'_T into C_T completes Lemma 1. This polynomial is not claimed to be optimal; its logarithm is all that the second-order proof uses.

Proof of Lemma 2. Choose $0 < m < M$ with $m\mathbf{I}_2 \preceq \mathbf{S} \preceq M\mathbf{I}_2$ on $\mathcal{K}$, and $0 < d < \min\{m, D_1, D_2\}$. The covariance $d\mathbf{I}_2$ is feasible for all nonnegative private rates. The determinant has a maximizer on the compact feasible set obtained by allowing $\mathbf{K} \succeq 0$, and

$$|\mathbf{K}^*| \ge d^2 > 0, \tag{277}$$

$$\lambda_{\min}(\mathbf{K}^*) = \frac{|\mathbf{K}^*|}{\lambda_{\max}(\mathbf{K}^*)} \ge \frac{d^2}{M}. \tag{278}$$

where Eq. (277) compares with $d\mathbf{I}_2$, and Eq. (278) uses the two-dimensional determinant and $\mathbf{K}^* \preceq M\mathbf{I}_2$. Thus the maximizer is positive definite and minimizes (27). Taking $\kappa \le \min\{d^2/M, 1/M\}$ proves the uniform claim. □

With these uniform covariance bounds established, recall $\mathbf{B} = (\mathbf{S}-\mathbf{K})^{1/2}$. Under the analysis distribution $Q_{\mathbf{S},\mathbf{K}}$, let $\mathbf{U} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_2)$ and $\mathbf{N} \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$ be independent, and set $\mathbf{X} = \mathbf{B}\mathbf{U} + \mathbf{N}$. Its source marginal is $Q_{\mathbf{S}}$; $Q_{\mathbf{U}}$ is nonsingular even when $\mathbf{W} = \mathbf{B}\mathbf{U}$ is singular. Since $\mathbf{X}$ depends on $\mathbf{U}$ only through $\mathbf{W}$, their mutual informations with $\mathbf{X}$ are equal. Gaussian conditioning gives

$$I_{\mathbf{S},\mathbf{K}} := I_Q(\mathbf{X};\mathbf{U}) = \frac{1}{2}\log\frac{|\mathbf{S}|}{|\mathbf{K}|}, \tag{279}$$

$$\begin{aligned} \imath_{\mathbf{S},\mathbf{K}}(\mathbf{u};\mathbf{x}) &:= \log\frac{dQ_{\mathbf{X}|\mathbf{U}=\mathbf{u}}}{dQ_{\mathbf{S}}}(\mathbf{x}) \\ &= I_{\mathbf{S},\mathbf{K}} - \tfrac{1}{2}(\mathbf{x}-\mathbf{B}\mathbf{u})^{\mathsf{T}}\mathbf{K}^{-1}(\mathbf{x}-\mathbf{B}\mathbf{u}) + \tfrac{1}{2}\mathbf{x}^{\mathsf{T}}\mathbf{S}^{-1}\mathbf{x}, \end{aligned} \tag{280}$$

$$\mathbf{U} \mid \mathbf{X} = \mathbf{x} \sim \mathcal{N}(\mathbf{B}^{\mathsf{T}}\mathbf{S}^{-1}\mathbf{x}, \mathbf{C}), \tag{281}$$

$$\mathbf{C} := \mathbf{I}_2 - \mathbf{B}^{\mathsf{T}}\mathbf{S}^{-1}\mathbf{B} = (\mathbf{I}_2 + \mathbf{B}^{\mathsf{T}}\mathbf{K}^{-1}\mathbf{B})^{-1} \succ 0. \tag{282}$$

where Eq. (280) divides the Gaussian densities and Eq. (282) is the Woodbury identity. All matrix coefficients are uniformly bounded under (141).

For a fixed shell block $\mathbf{x}^n$, draw $\mathbf{U}_i$ independently from these posterior kernels and write $Q(\cdot \mid \mathbf{x}^n)$ for their product. Define

$$J_n := \sum_{i\in[1:n]} \imath_{\mathbf{S},\mathbf{K}}(\mathbf{U}_i;\mathbf{x}_i), \tag{283}$$

$$E_{j,n} := \sum_{i\in[1:n]} ([\mathbf{x}_i]_j - (\mathbf{B}\mathbf{U}_i)_j)^2, \qquad j \in [1:2]. \tag{284}$$

The residual $\mathbf{N}_i = \mathbf{x}_i - \mathbf{B}\mathbf{U}_i$ has posterior mean $\mathbf{K}\mathbf{S}^{-1}\mathbf{x}_i$ and covariance $\mathbf{K} - \mathbf{K}\mathbf{S}^{-1}\mathbf{K}$. Summing its second moments and substituting (280) yields

$$\mathsf{E}_Q[E_{j,n} \mid \mathbf{x}^n] = nK_{jj} + n\mathbf{e}_j^{\mathsf{T}}\mathbf{K}\mathbf{S}^{-1}(\hat{\mathbf{S}}(\mathbf{x}^n) - \mathbf{S})\mathbf{S}^{-1}\mathbf{K}\mathbf{e}_j, \tag{285}$$

$$\mathsf{E}_Q[J_n \mid \mathbf{x}^n] = nI_{\mathbf{S},\mathbf{K}} + \frac{n}{2}\operatorname{tr}\left[\mathbf{S}^{-1}(\mathbf{S}-\mathbf{K})\mathbf{S}^{-1}(\hat{\mathbf{S}}(\mathbf{x}^n) - \mathbf{S})\right], \tag{286}$$

$$|\mathsf{E}_Q[J_n \mid \mathbf{x}^n] - nI_{\mathbf{S},\mathbf{K}}| + \sum_{j\in[1:2]} |\mathsf{E}_Q[E_{j,n} \mid \mathbf{x}^n] - nK_{jj}| \le Cn\left\|\hat{\mathbf{S}}(\mathbf{x}^n) - \mathbf{S}\right\| \le C'. \tag{287}$$

where the first two identities use the empirical second-moment matrix and cyclicity of trace, and Eq. (287) uses the uniform matrix bounds and shell width C_0/n. These are the conditional centers needed by the common covering argument.

B. Orthogonal Invariance and Uniform Small-Ball Control

Lemma 9 (Orthogonal invariance)**.** *Fix* $(\mathbf{S},\mathbf{K})$ *as above. For every* $\mathbf{x}^n \in \mathbb{R}^{2\times n}$ *and every orthogonal* $\mathbf{O} \in \mathbb{R}^{n\times n}$*, the posterior distribution of* $(J_n, E_{1,n}, E_{2,n})$ *is the same for* $\mathbf{x}^n$ *and* $\mathbf{x}^n\mathbf{O}$*. Hence that distribution depends on the block only through* $\mathbf{x}^n(\mathbf{x}^n)^\mathsf{T}$.

Proof. Let $\mathbf{Z}^n$ have independent standard Gaussian columns. The posterior can be represented as

$$\mathbf{U}^n(\mathbf{x}^n) = \mathbf{B}^\mathsf{T}\mathbf{S}^{-1}\mathbf{x}^n + \mathbf{C}^{1/2}\mathbf{Z}^n, \tag{288}$$

$$\mathbf{U}^n(\mathbf{x}^n\mathbf{O}) \overset{d}{=} \mathbf{B}^\mathsf{T}\mathbf{S}^{-1}\mathbf{x}^n\mathbf{O} + \mathbf{C}^{1/2}\mathbf{Z}^n\mathbf{O} = \mathbf{U}^n(\mathbf{x}^n)\mathbf{O}. \tag{289}$$

where Eq. (289) uses $\mathbf{Z}^n\mathbf{O} \overset{d}{=} \mathbf{Z}^n$. Thus both the source and residual row Gram matrices are jointly invariant. Their residual diagonal entries are $E_{1,n}, E_{2,n}$, while

$$J_n = nI_{\mathbf{S},\mathbf{K}} - \tfrac{1}{2}\operatorname{tr}[\mathbf{K}^{-1}(\mathbf{x}^n - \mathbf{B}\mathbf{U}^n)(\mathbf{x}^n - \mathbf{B}\mathbf{U}^n)^\mathsf{T}] + \tfrac{1}{2}\operatorname{tr}[\mathbf{S}^{-1}\mathbf{x}^n(\mathbf{x}^n)^\mathsf{T}]. \tag{290}$$

Equation (290) follows by summing (280), proving joint invariance of all three statistics. Blocks with the same row Gram matrix are related by a right orthogonal transformation: the Gram-preserving isometry of their row spaces extends to $\mathbb{R}^n$. □

For $n \geq 3$, define $\mathbf{H}_n \in \mathbb{R}^{2\times n}$ to have ith column $\sqrt{2/n}(\cos(2\pi i/n), \sin(2\pi i/n))^\mathsf{T}$. The trigonometric sums give $\mathbf{H}_n\mathbf{H}_n^\mathsf{T} = \mathbf{I}_2$. With $\hat{\mathbf{S}} = \hat{\mathbf{S}}(\mathbf{x}^n)$, the bounded representative satisfies

$$\mathbf{x}_\circ^n := \sqrt{n}\,\hat{\mathbf{S}}^{1/2}\mathbf{H}_n, \tag{291}$$

$$\hat{\mathbf{S}}(\mathbf{x}_\circ^n) = \hat{\mathbf{S}}^{1/2}\mathbf{H}_n\mathbf{H}_n^\mathsf{T}\hat{\mathbf{S}}^{1/2} = \hat{\mathbf{S}}, \tag{292}$$

$$\max_{i\in[1:n]} \|\mathbf{x}_{\circ,i}\| \leq \sqrt{2\lambda_{\max}(\hat{\mathbf{S}})} \leq C. \tag{293}$$

where the last bound uses the compact shell family. Lemma 9 therefore permits conditional aggregate probabilities to be evaluated at this representative even if the original block has large individual coordinates.

By Lemma 9, replace the original block by $\mathbf{x}_\circ^n$ and relabel this representative as $\mathbf{x}^n$ for the following conditional calculation. In particular, its columns satisfy (293). Define the centered vectors

$$\mathbf{T}_i := \begin{bmatrix} \imath_{\mathbf{S},\mathbf{K}}(\mathbf{U}_i;\mathbf{x}_i) \\ ([\mathbf{x}_i]_1 - (\mathbf{B}\mathbf{U}_i)_1)^2 \\ ([\mathbf{x}_i]_2 - (\mathbf{B}\mathbf{U}_i)_2)^2 \end{bmatrix} - \mathsf{E}_Q\left[\left.\begin{bmatrix} \imath_{\mathbf{S},\mathbf{K}}(\mathbf{U}_i;\mathbf{x}_i) \\ ([\mathbf{x}_i]_1 - (\mathbf{B}\mathbf{U}_i)_1)^2 \\ ([\mathbf{x}_i]_2 - (\mathbf{B}\mathbf{U}_i)_2)^2 \end{bmatrix}\right| \mathbf{x}_i\right] \in \mathbb{R}^3. \tag{294}$$

These vectors are independent, not necessarily identically distributed. At the bounded representative, they are quadratic polynomials of standard Gaussian innovations with uniformly bounded coefficients. Consequently,

$$\sup_{i\in[1:n]} \mathsf{E}_Q[\|\mathbf{T}_i\|^3 \mid \mathbf{x}_i] \leq C. \tag{295}$$

Lemma 10 (Uniform conditional small ball)**.** *There are* $c > 0$ *and* $n_0 \in \mathbb{N}$ *such that, for all* $n \geq n_0$*,*

$$Q\left\{\left\|\sum_{i\in[1:n]} \mathbf{T}_i\right\| \leq \frac{\sqrt{n}}{64\log n} \,\middle|\, \mathbf{x}^n\right\} \geq \frac{c}{(\log n)^3} \tag{296}$$

uniformly over (141) *and* $\mathbf{x}^n \in \mathcal{T}_n(\mathbf{S})$.

Proof. The argument separates variance directions: after whitening, the large-variance subspace is handled by a multivariate Berry–Esseen bound, whereas the small-variance subspace stays inside the target window with high probability by a second-moment estimate. By Lemma 9, use the bounded representative. Let $\mathbf{\Gamma}_n = \sum_{i\in[1:n]} \mathsf{Cov}_Q(\mathbf{T}_i \mid \mathbf{x}_i)$, the covariance of the centered sum. Equation (295) gives $\lambda_{\max}(\mathbf{\Gamma}_n) \leq \operatorname{tr}(\mathbf{\Gamma}_n) \leq Cn$. Set

$$\tau_n := n/(\log n)^8, \tag{297}$$

$$a_n := \sqrt{n}/(64\log n). \tag{298}$$

Let $\mathbf{P}_\mathrm{L}$ and $\mathbf{P}_\mathrm{S}$ project onto eigenvalues at least τ_n and below it. Markov's inequality gives

$$\begin{aligned} Q\left\{\left\|\mathbf{P}_\mathrm{S}\sum_{i\in[1:n]} \mathbf{T}_i\right\| > a_n/2 \,\middle|\, \mathbf{x}^n\right\} &\leq \frac{4\operatorname{tr}(\mathbf{P}_\mathrm{S}\mathbf{\Gamma}_n)}{a_n^2} \\ &\leq \frac{12\tau_n}{a_n^2} \leq \frac{C}{(\log n)^6}. \end{aligned} \tag{299}$$

where the trace contains at most three eigenvalues smaller than τ_n. If $q = \operatorname{rank}(\mathbf{P}_\mathrm{L}) = 0$, this proves the lemma.

Otherwise choose an orthonormal basis matrix $\mathbf{O}_{\mathrm{L}} \in \mathbb{R}^{3\times q}$ and define $\tilde{\mathbf{\Gamma}}_{\mathrm{L}} = \mathbf{O}_{\mathrm{L}}^{\mathsf{T}}\mathbf{\Gamma}_n\mathbf{O}_{\mathrm{L}}$ and $\mathbf{Z}_i' = \tilde{\mathbf{\Gamma}}_{\mathrm{L}}^{-1/2}\mathbf{O}_{\mathrm{L}}^{\mathsf{T}}\mathbf{T}_i$. Then

$$\mathsf{Cov}_Q\left(\sum_{i\in[1:n]}\mathbf{Z}_i' \,\middle|\, \mathbf{x}^n\right) = \mathbf{I}_q, \tag{300}$$

$$\begin{aligned}\sum_{i\in[1:n]}\mathsf{E}_Q[\|\mathbf{Z}_i'\|^3 \mid \mathbf{x}^n] &\le \tau_n^{-3/2}\sum_{i\in[1:n]}\mathsf{E}_Q[\|\mathbf{T}_i\|^3 \mid \mathbf{x}_i]\\ &\le Cn\tau_n^{-3/2} = C(\log n)^{12}/\sqrt{n}.\end{aligned} \tag{301}$$

where whitening gives the identity covariance, and the moment bound uses $\lambda_{\min}(\tilde{\mathbf{\Gamma}}_{\mathrm{L}}) \ge \tau_n$. Bentkus' theorem for independent summands [25] therefore bounds the error on every convex Borel set by the last expression, uniformly since $q \le 3$.

Choose $d_0 > 0$ independent of the parameters so that the inverse whitening map sends a ball of radius $d_0/\log n$ into one of radius $a_n/2$. This is possible because $\lambda_{\max}(\tilde{\mathbf{\Gamma}}_{\mathrm{L}}) \le Cn$. A standard q-dimensional Gaussian $\mathbf{N}_q$ assigns that ball probability at least $c(\log n)^{-q} \ge c(\log n)^{-3}$, by a lower bound on its density in the unit ball. Hence

$$\begin{aligned}Q\left\{\left\|\mathbf{P}_{\mathrm{L}}\sum_{i\in[1:n]}\mathbf{T}_i\right\| \le a_n/2 \,\middle|\, \mathbf{x}^n\right\} &\ge \Pr\{\|\mathbf{N}_q\| \le d_0/\log n\} - C(\log n)^{12}/\sqrt{n}\\ &\ge c'(\log n)^{-3},\end{aligned} \tag{302}$$

$$\begin{aligned}Q\left\{\left\|\sum_{i\in[1:n]}\mathbf{T}_i\right\| \le a_n \,\middle|\, \mathbf{x}^n\right\} &\ge c'(\log n)^{-3} - C(\log n)^{-6}\\ &\ge c''(\log n)^{-3}.\end{aligned} \tag{303}$$

where Eq. (302) applies the convex-set Berry–Esseen estimate and $(\log n)^{12}/\sqrt{n} = o((\log n)^{-3})$. Equation (303) uses the triangle inequality and subtracts the small-component failure bound (299). No independence between the projections is required. □

C. Whole-Shell Covering and Code Assembly

For a fixed shell block, define the analysis-only set

$$\mathcal{G}_n(\mathbf{x}^n) := \left\{\mathbf{u}^n \in \mathbb{R}^{2\times n} :\ J_n(\mathbf{u}^n;\mathbf{x}^n) \le nI_{\mathbf{S},\mathbf{K}} + s_n/16,\ E_{j,n}(\mathbf{u}^n;\mathbf{x}^n) \le nK_{jj} + s_n/16,\ j\in[1:2]\right\}. \tag{304}$$

The bounded mean shifts in (287) and Lemma 10 give $Q\{\mathcal{G}_n(\mathbf{x}^n) \mid \mathbf{x}^n\} \ge c/(\log n)^3$ for large n. The posterior and $Q_{\mathbf{U}}^n$ are mutually absolutely continuous by (282). Bayes' formula therefore yields

$$\begin{aligned}Q_{\mathbf{U}}^n\{\mathcal{G}_n(\mathbf{x}^n)\} &= \mathsf{E}_Q[e^{-J_n}\mathbb{1}_{\mathcal{G}_n(\mathbf{x}^n)} \mid \mathbf{x}^n]\\ &\ge e^{-nI_{\mathbf{S},\mathbf{K}}-s_n/16}Q\{\mathcal{G}_n(\mathbf{x}^n) \mid \mathbf{x}^n\}\\ &\ge \frac{c}{(\log n)^3}e^{-nI_{\mathbf{S},\mathbf{K}}-s_n/16} =: p_n.\end{aligned} \tag{305}$$

where Eq. (305) uses $dQ_{\mathbf{U}}^n/dQ_{\mathbf{U}^n|\mathbf{x}^n} = e^{-J_n}$, the information-density cutoff, and the conditional small-ball bound.

For the independent codebook draws in (110), let $\mathcal{E}_n^{(0)}(\mathbf{x}^n) := \bigcap_{m_0\in[1:M_0]}\{\mathbf{u}^n(m_0) \notin \mathcal{G}_n(\mathbf{x}^n)\}$. Take a maximal δ_n-separated subset $\mathcal{N}_n$ of the shell, with $\delta_n = 1/\log n$. It is a finite net with centers in the shell. Since the shell lies in a radius-$C\sqrt{n}$ ball in $\mathbb{R}^{2n}$, a disjoint-ball volume comparison gives

$$|\mathcal{N}_n| \le (1 + C\sqrt{n}\log n)^{2n}, \tag{306}$$

$$Q_{0,n}(\mathcal{E}_n^{(0)}(\mathbf{x}^n)) \le (1-p_n)^{M_0} \le e^{-M_0p_n} \le \exp\{-c(\log n)^{-3}e^{s_n/16}\}. \tag{307}$$

where Eq. (307) uses independence and the common codebook size (107). For $\mathcal{E}_n^{(1)} := \bigcup_{\mathbf{x}^n\in\mathcal{N}_n}\mathcal{E}_n^{(0)}(\mathbf{x}^n)$,

$$\begin{aligned}Q_{0,n}(\mathcal{E}_n^{(1)}) &\le |\mathcal{N}_n|\exp\{-c(\log n)^{-3}e^{s_n/16}\}\\ &\le \exp\{Cn\log n - c(\log n)^{-3}e^{s_n/16}\} \longrightarrow 0.\end{aligned} \tag{308}$$

where Eq. (308) applies the union bound and $e^{s_n/16}/((\log n)^3 n\log n) \to \infty$. Fix a common codebook good for every net point.

For an arbitrary shell block $\mathbf{x}^n$, take a nearest net point $\bar{\mathbf{x}}^n$ and a codeword good for it. Then, for $j \in [1:2]$,

$$\begin{aligned}\|(\mathbf{x}^n)_j - (\mathbf{B}\mathbf{u}^n)_j\|^2 &\le \left(\sqrt{nK_{jj} + s_n/16} + \delta_n\right)^2 \\ &\le nK_{jj} + \chi_\kappa s_n = \xi_{j,n}^2, \end{aligned} \tag{309}$$

where Eq. (309) uses the triangle inequality, $\delta_n\sqrt{n} = s_n$, $K_{jj} \le \kappa^{-1}$, and the fixed margin (111). Only the residual test, not the information-density test, is needed away from net points. The previously specified private coverings therefore give both distortions for the encoder and decoders of (118)–(119).

Their cardinalities from Lemma 1 obey

$$\begin{aligned}\log M_j &\le \log C_{\mathrm{cov}} + 4\log n + \frac{n}{2}\left[\log\frac{K_{jj} + \chi_\kappa s_n/n}{D_j}\right]^+ \\ &\le \frac{n}{2}[\log(K_{jj}/D_j)]^+ + \frac{n}{2}\log\left(1 + \frac{\chi_\kappa s_n}{nK_{jj}}\right) + C\log n \\ &\le \frac{n}{2}[\log(K_{jj}/D_j)]^+ + Cs_n + C\log n. \end{aligned} \tag{310}$$

where Eq. (310) uses the positive-part inequality, $\log(1+x) \le x$, and $K_{jj} \ge \kappa$. Together with (107), this proves Lemma 3. All constants are uniform, including when $\mathbf{S} - \mathbf{K}$ loses rank.

Proof of Lemma 4. Use the construction in Section IV-A. Lemmas 2 and 3 provide one uniform κ and covering codebooks for every retained cell. The empty retained set uses the specified fallback. Otherwise,

$$M_0 = 1 + \sum_{q\in\mathcal{J}_n} M_{0,q} \le 1 + N_n e^{nr_0 + Cs_n} \le e^{nr_0 + Cs_n + C\log n}, \tag{311}$$

$$M_j = \max_{q\in\mathcal{J}_n} M_{j,q} \le e^{nr_j + Cs_n + C\log n}, \qquad j \in [1:2]. \tag{312}$$

where the first line uses the disjoint common alphabets, $N_n \le Cn^3$, and (142); the second uses private padding and (143). The common message identifies the cell, so neither private size acquires a cell factor. Integer rounding adds only a constant.

The finite smallest-index and grid tests are Borel. Each retained cell lies in a covered shell, whence

$$\begin{aligned}\varepsilon_n &\le P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \notin \mathcal{K}_0\} + \sum_{q\in[1:N_n]\setminus\mathcal{J}_n} P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \in \mathcal{K}_0, q_n(\hat{\mathbf{\Sigma}}_n) = \mathbf{S}_q\} \\ &= P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \notin \mathcal{K}_0\} + P_{XY}^n\{\hat{\mathbf{\Sigma}}_n \in \mathcal{K}_0, \mathsf{R}_0(\mathbf{S}_n; r_1, r_2) > r_0\}. \end{aligned} \tag{313}$$

where Eq. (313) uses the partition and definition of $\mathcal{J}_n$. Enlarging the common constant proves both conclusions. □

APPENDIX D
GAUSSIAN CELL AND IMAGE-SIZE ESTIMATES

We prove Lemma 8. For a fixed code and $m_0 \in [1 : M_{0,n}]$, define

$$\mathcal{A}_{n,m_0} := \mathcal{A}_n \cap f_0^{-1}(\{m_0\}), \tag{314}$$

$$\mathcal{A}_{j,m_0} := \bigcup_{m_j\in[1:M_{j,n}]} \mathcal{B}_n(\phi_j(m_0,m_j), \sqrt{nD_j}), \qquad j \in [1:2], \tag{315}$$

$$\mathcal{S}_n(s) := \{x^n \in \mathbb{R}^n : |n^{-1}\|x^n\|^2 - s| \le 1/(2n)\}. \tag{316}$$

The first set fixes the common index of successful blocks; the second is the private decoding image. Membership in $\mathcal{C}_n(\mathbf{S})$ requires the jth row to lie in $\mathcal{S}_n(s_{jj})$.

Lemma 11 (Gaussian cell geometry)**.** *For a compact $\mathcal{K} \subset \mathbb{S}_{++}^2$, there are constants $(c, C) \in \mathbb{R}_+^2$ with $\min\{c, C\} > 0$ and $n_0 \in \mathbb{N}$ such that, for $n \ge n_0$ and every $\mathbf{S} \in \mathcal{K}$,*

$$Q_{\mathbf{S}}^n\{\mathcal{C}_n(\mathbf{S})\} \ge cn^{-3/2}, \tag{317}$$

$$\sup_{(\mathbf{x}^n,\mathbf{y}^n)\in\mathcal{C}_n(\mathbf{S})^2} \left|\log\frac{dP_{XY}^n}{dQ_{\mathbf{S}}^n}(\mathbf{x}^n) - \log\frac{dP_{XY}^n}{dQ_{\mathbf{S}}^n}(\mathbf{y}^n)\right| \le C. \tag{318}$$

Under $Q_{\mathbf{S}}^n$, the matrix $\mathbf{Y} = n\hat{\mathbf{S}}(\mathbf{X}^n)$ has Wishart density

$$f_{n,\mathbf{S}}(\mathbf{Y}) = \frac{|\mathbf{Y}|^{(n-3)/2} e^{-\operatorname{tr}(\mathbf{S}^{-1}\mathbf{Y})/2}}{2^n |\mathbf{S}|^{n/2} \Gamma_2(n/2)}, \qquad \mathbf{Y} \in \mathbb{S}_{++}^2, \tag{319}$$

with respect to its three independent entries, where $\Gamma_2(a) = \sqrt{\pi}\Gamma(a)\Gamma(a-1/2)$. For bounded symmetric $\mathbf{H}$ and $\mathbf{S} \in \mathcal{K}$,

$$\log|n\mathbf{S} + \mathbf{H}| = 2\log n + \log|\mathbf{S}| + n^{-1}\operatorname{tr}(\mathbf{S}^{-1}\mathbf{H}) + O(n^{-2}), \tag{320}$$

$$\log f_{n,\mathbf{S}}(n\mathbf{S} + \mathbf{H}) = -\tfrac{3}{2}\log n + O(1). \tag{321}$$

where Eq. (320) is the uniform log-determinant expansion and Eq. (321) substitutes it into the density and applies Stirling's formula. Compact positive definiteness makes the constants uniform and keeps $n\mathbf{S} + \mathbf{H}$ positive definite for large n. In these coordinates the cell contains a fixed positive-volume box of $\mathbf{H}$ values. Integrating the density lower bound over this box gives (317); half-open boundaries have zero mass.

For the true source, division of Gaussian densities gives

$$\log \frac{dP_{XY}^n}{dQ_{\mathbf{S}}^n}(\mathbf{x}^n) = \frac{n}{2}\log\frac{|\mathbf{S}|}{|\mathbf{\Sigma}|} - \frac{n}{2}\operatorname{tr}[(\mathbf{\Sigma}^{-1} - \mathbf{S}^{-1})\hat{\mathbf{S}}(\mathbf{x}^n)], \tag{322}$$

$$\left|\log \frac{dP_{XY}^n}{dQ_{\mathbf{S}}^n}(\mathbf{x}^n) - \log \frac{dP_{XY}^n}{dQ_{\mathbf{S}}^n}(\mathbf{y}^n)\right| \le \frac{n}{2}\left\|\mathbf{\Sigma}^{-1} - \mathbf{S}^{-1}\right\| \left\|\hat{\mathbf{S}}(\mathbf{x}^n) - \hat{\mathbf{S}}(\mathbf{y}^n)\right\| \le C. \tag{323}$$

where the last inequality holds for $(\mathbf{x}^n, \mathbf{y}^n) \in \mathcal{C}_n(\mathbf{S})^2$ by the $O(1/n)$ cell diameter and compactness. This proves (318).

Lemma 12 (Marginal image-size estimate)**.** *Fix $D > 0$ and a compact interval $\mathcal{I} \subset (D, \infty)$. For some $C < \infty$ and all sufficiently large n,*

$$\mathcal{N}(0, s\mathbf{I}_n)\big(\mathcal{B}_n(\hat{x}^n, \sqrt{nD}) \cap \mathcal{S}_n(s)\big) \le n^C \exp\{-\tfrac{n}{2}\log(s/D)\} \tag{324}$$

uniformly for $s \in \mathcal{I}$ and every center $\hat{x}^n \in \mathbb{R}^n$.

Write a vector under $\mathcal{N}(0, s\mathbf{I}_n)$ as $X^n = R\mathbf{\Theta}$, where R and the uniform direction $\mathbf{\Theta}$ are independent. For a center of norm $\sqrt{n}\,t$ and conditional power $u = R^2/n > D$, the distortion ball restricts the angle by

$$\cos\theta \ge \frac{u + t^2 - D}{2t\sqrt{u}}, \qquad t > 0, \tag{325}$$

$$\min_{t>0} \frac{u + t^2 - D}{2t\sqrt{u}} = \sqrt{1 - D/u}. \tag{326}$$

where the second identity minimizes at $t^2 = u - D$. For $t = 0$ or a cutoff exceeding one, the intersection is empty. The first coordinate of $\mathbf{\Theta}$ has density $\omega_n(1-v^2)^{(n-3)/2}$, with $\omega_n = \Gamma(n/2)/(\sqrt{\pi}\Gamma((n-1)/2)) \le C\sqrt{n}$. For $z > 0$,

$$\begin{aligned}\int_z^1 (1-v^2)^{(n-3)/2}\,dv &= \frac{1}{2}\int_0^{1-z^2} w^{(n-3)/2}(1-w)^{-1/2}\,dw \\ &\le \frac{(1-z^2)^{(n-1)/2}}{z(n-1)}.\end{aligned} \tag{327}$$

where Eq. (327) substitutes $w = 1 - v^2$ and uses $\sqrt{1-w} \ge z$. On the power shell, $u = s + O(1/n)$ uniformly; compactness of $\mathcal{I} \subset (D, \infty)$ keeps $z = \sqrt{1 - D/u}$ bounded away from zero. Hence

$$\begin{aligned}\Pr\{X^n \in \mathcal{B}_n(\hat{x}^n, \sqrt{nD}) \mid R\} &\le Cn^{-1/2}(D/u)^{(n-1)/2} \\ &\le C'(D/s)^{n/2}.\end{aligned} \tag{328}$$

where Eq. (328) uses the cap bound and $(s/u)^{(n-1)/2} = O(1)$ on the thin shell. Integrating over the shell radii, whose probability is at most one, proves (324) uniformly in s and the center.

We now combine the two estimates to prove Lemma 8. Fix a code, $\mathbf{S} \in \mathcal{K}$, and n sufficiently large. Put $s_j = s_{jj}$ and $Q_{\mathbf{S},j} = \mathcal{N}(0, s_j)$. By the union bound and Lemma 12,

$$\begin{aligned}Q_{\mathbf{S},j}^n(\mathcal{A}_{j,m_0} \cap \mathcal{S}_n(s_j)) &\le \min\{1, M_{j,n} n^C e^{-\frac{n}{2}\log(s_j/D_j)}\} \\ &\le e^{-n\delta_{j,n}(\mathbf{S}) + C\log n}, \qquad j \in [1:2].\end{aligned} \tag{329}$$

where Eq. (329) uses the definition (190), including zero deficits.

Let $\rho_{\mathbf{S}} = \rho(\mathbf{S})$ and let $P_{\mathrm{G},n} := \mathcal{N}(\mathbf{0}, \mathbf{I}_n)$ be the standard Gaussian distribution on $\mathbb{R}^n$. For a measurable function $h : \mathbb{R}^n \to \mathbb{R}$ and $p > 1$, write $\|h\|_p := (\int |h|^p \, dP_{\mathrm{G},n})^{1/p}$. Define the Ornstein–Uhlenbeck operator by

$$(T_{\rho_{\mathbf{S}}} h)(\mathbf{x}) := \int h\big(\rho_{\mathbf{S}}\mathbf{x} + \sqrt{1-\rho_{\mathbf{S}}^2}\,\mathbf{z}\big)\, P_{\mathrm{G},n}(d\mathbf{z}), \qquad \mathbf{x} \in \mathbb{R}^n. \tag{330}$$

For Borel sets $\mathcal{A}_1, \mathcal{A}_2 \subset \mathbb{R}^n$, set $h_j(\mathbf{x}) := \mathbb{1}_{\mathcal{A}_j}(\sqrt{s_j}\,\mathbf{x})$, $j \in [1:2]$. Gaussian conditioning identifies $T_{\rho_{\mathbf{S}}} h_2$ with the conditional expectation of the second standardized indicator given the first standardized source. If $(p, q) \in \mathcal{H}_{\rho_{\mathbf{S}}}$ and $p' = p/(p-1)$, Nelson's Gaussian hypercontractive inequality [26, Theorem 3], with the dimension extension described in [27, Corollary 4.1 and the proof of Corollary 4.2], gives

$$\|T_{\rho_{\mathbf{S}}} h\|_{p'} \le \|h\|_q \qquad \text{if } 1 < q \le p' < \infty \text{ and } \rho_{\mathbf{S}}^2 \le \frac{q-1}{p'-1}. \tag{331}$$

The exponent condition is precisely $(p-1)(q-1) \ge \rho_{\mathbf{S}}^2$, and the constant is one in every dimension n. When $q > p'$, the same norm inequality follows from contraction in $L^q(P_{\mathrm{G},n})$ and the probability-space embedding into $L^{p'}(P_{\mathrm{G},n})$. Consequently,

$$\begin{aligned} Q_{\mathbf{S}}^n(\mathcal{A}_1 \times \mathcal{A}_2) &= \mathsf{E}_{P_{\mathrm{G},n}}[h_1 T_{\rho_{\mathbf{S}}} h_2] \le \|h_1\|_p \, \|T_{\rho_{\mathbf{S}}} h_2\|_{p'} \\ &\le \|h_1\|_p \, \|h_2\|_q = Q_{\mathbf{S},1}^n(\mathcal{A}_1)^{1/p} Q_{\mathbf{S},2}^n(\mathcal{A}_2)^{1/q}. \end{aligned} \tag{332}$$

where the first inequality is Hölder's and the second uses (331) or the preceding contraction argument. Endpoint contributions are obtained by limits, never by setting $p = 1$.

Successful decoding with common index m_0 inside a cell implies membership in both marginal images and power shells. Therefore

$$\begin{aligned} Q_{\mathbf{S}}^n(\mathcal{A}_{n,m_0} \cap \mathcal{C}_n(\mathbf{S})) &\le Q_{\mathbf{S}}^n\big((\mathcal{A}_{1,m_0} \cap \mathcal{S}_n(s_1)) \times (\mathcal{A}_{2,m_0} \cap \mathcal{S}_n(s_2))\big) \\ &\le \inf_{(p,q)\in\mathcal{H}_{\rho_{\mathbf{S}}}} e^{-n(\delta_{1,n}(\mathbf{S})/p + \delta_{2,n}(\mathbf{S})/q) + C\log n} \\ &= e^{-n\psi_{\rho_{\mathbf{S}}}(\delta_{1,n}(\mathbf{S}),\delta_{2,n}(\mathbf{S})) + C\log n}. \end{aligned} \tag{333}$$

where Eq. (333) combines (329) and (332), then uses (60). The polynomial factor is uniform before optimization since $1/p + 1/q \le 2$. Summing this bound over $M_{0,n}$ common messages gives

$$Q_{\mathbf{S}}^n(\mathcal{A}_n \cap \mathcal{C}_n(\mathbf{S})) \le e^{\log M_{0,n} - n\psi_{\rho_{\mathbf{S}}}(\delta_{1,n}(\mathbf{S}),\delta_{2,n}(\mathbf{S})) + C\log n}. \tag{334}$$

No separation of the common encoder across sources has been assumed.

Set $\mathcal{C} = \mathcal{C}_n(\mathbf{S})$ and $\ell = dP_{XY}^n / dQ_{\mathbf{S}}^n$. The cell has positive probability by Lemma 11, so

$$\begin{aligned} P_{XY}^n(\mathcal{A}_n \mid \mathcal{C}) &= \frac{\int_{\mathcal{A}_n \cap \mathcal{C}} \ell \, dQ_{\mathbf{S}}^n}{\int_{\mathcal{C}} \ell \, dQ_{\mathbf{S}}^n} \\ &\le \frac{\sup_{\mathcal{C}} \ell}{\inf_{\mathcal{C}} \ell} \frac{Q_{\mathbf{S}}^n(\mathcal{A}_n \cap \mathcal{C})}{Q_{\mathbf{S}}^n(\mathcal{C})} \\ &\le C n^k e^{\log M_{0,n} - n\psi_{\rho(\mathbf{S})}(\delta_{1,n}(\mathbf{S}),\delta_{2,n}(\mathbf{S}))}. \end{aligned} \tag{335}$$

where Eq. (335) bounds the conditional likelihood ratio and applies (318), (317), and (334). Taking the minimum with one proves Lemma 8. Its constants were fixed independently of the code, rates, centers, and selected covariance.

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